Text properties and children's reading pace in a multilingual storytelling app

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Abstract

Children increasingly read machine-generated stories on screens, and whether the text properties that govern reading under controlled conditions govern it there is unknown. We will analyse logs from *Let's Story*, a feature of the *Applaydu* app, in which children aged 7 and older read in ten or more languages. We will model reading time per word from lexical difficulty, grammatical load and referential continuity. Six confirmatory hypotheses concern these associations, their moderation by age band, narration's selectivity, children's difficulty ratings and accumulated in-app reading. Bayesian crossed-effects models will represent readings within texts, children and languages. Coefficients outside a region of practical equivalence would support the hypotheses, in either direction for the two that adjudicate between rival developmental accounts, and posteriors inside it would show that the laboratory effects do not transfer. Because families choose narration, texts and continued use, we will report every coefficient as a conditional association.

Introduction

A generation of children now does part of its reading on a screen, and some of what appears there was written by a machine. In the United Kingdom, one in five 8- to 18-year-olds read fiction on screen at least monthly in 2025, print fiction had fallen from three in five to one in two since 2017, and the children who read fiction only on screen were disproportionately those who enjoy reading least (Picton & Clark, 2026). Commercial storytelling applications generate a narrative to order, read it aloud in the family's language, and record what the child did with it. One such feature reported 2.4 million story sessions in its first six months (Ratner et al., 2025), while the parents interviewed in one small study were mostly uncomfortable with a machine writing the words of a child's story (Jin & Yuan, 2025). The behavioural record such platforms accumulate is far larger than any laboratory corpus of children's reading, spans ten or more languages, and arises under exactly the conditions that laboratory work has to strip away. We ask whether the text properties that govern reading under controlled conditions still govern it outside the laboratory, and whether their hold on the reader changes over the years in which decoding becomes automatic. Whatever the answer, the field gains an estimate it does not have: how far three determinants of reading established under laboratory control carry into unsupervised reading of generated stories, at scale, in ten languages, with the developmental change in each.

The laboratory answer is well established at three levels of linguistic description. At the level of the word, high-frequency words are recognised faster and fixated for less time than low-frequency words, and the effects of frequency and of word length on text reading are closely similar across 12 alphabetic languages (Kuperman et al., 2024). Word-length effects have been shown to take a script-specific form outside the Roman alphabet (Lahoud et al., 2023), and Korean is the one language here to which that caution applies. Children's eye movements are driven by the frequency of the words they meet (H. S. S. L. Joseph et al., 2013), and in the one study here that tested children against adult controls, the length and frequency effects were the larger in the children (Tiffin-Richards & Schroeder, 2015). The difference is what an account on which that effect tracks accumulated exposure predicts (Brysbaert et al., 2018). The same developmental pattern appears outside alphabetic scripts (Lian et al., 2026). The Lexical Quality Hypothesis gives that difference a description (Perfetti, 2007; Perfetti & Stafura, 2014). A word is retrieved quickly and reliably in proportion to how well specified and tightly bound its orthographic, phonological and semantic constituents are, and that quality is built by reading itself (Castles et al., 2018; Nation, 2017). On that account, a text's lexical properties, the age band and a child's accumulated reading bear on the cost of identifying a word, so Hypotheses 1 and 6 test that account at different grains.

A second source of word-level cost is contextual. On surprisal theory, a word costs time in proportion to how unexpected it is given what precedes it (Hale, 2001; Levy, 2008), and the relation is close to linear in surprisal (Shain et al., 2024; Smith & Levy, 2013) and holds in eleven languages from five language families (Wilcox et al., 2023). Linearity at the word level makes a text's mean per-word surprisal the natural text-level summary of that cost, which this design uses. A text's overall difficulty moderates how strongly per-word surprisal bears on reading time, so the text-level mean is not a neutral summary of a constant word-level effect (Chen et al., 2026). Even the separability of frequency from predictability depends on the language model estimating the latter, since dissociable frequency effects shrink as surprisal predictors improve (Oh & Schuler, 2025). That qualifies the reported additivity of the two in naturalistic reading (Shain, 2024). A component that pools

them indexes their joint contribution without resolving it, which is what the confirmatory claim asserts. How subword tokens are aggregated to words has a documented effect on psycholinguistic estimates (Giulianelli et al., 2024), and the unit over which surprisal should be defined is itself under revision (Kiegeland et al., 2026), which is why the aggregation rule is fixed under Materials. In children, contextual facilitation appears in the later eye-movement measures that a whole-text duration absorbs (Tiffin-Richards & Schroeder, 2020).

Related lexical properties behave similarly. Concrete words are fixated for less time than abstract ones in natural reading, on every fixation measure where length and frequency were matched (Yao et al., 2024) and on first-fixation duration where they were covaried instead (Magnabosco & Hauk, 2024). The advantage is usually traced to perceptual grounding, on accounts that now treat conceptual processing as partly modal and partly amodal (Kaup et al., 2025). The age at which a word was learned bears on how quickly it is identified over and above how often it occurs (Juhasz, 2005), which is why age of acquisition is carried in the lexical family. Lexical density, the proportion of tokens that are content words, is carried for a different reason. Function words are skipped far more often than content words (Rayner, 1998), so a text of higher density presents more tokens that must be fixated and identified. A text of low lexical diversity repeats its word types, and reading times for a word fall over repeated encounters, in children of 10 and 11 reading sentences (H. Joseph & Nation, 2018), so a diversity index (McCarthy & Jarvis, 2010) tracks the share of tokens a reader meets for the first time. A word-level measure of informational cost predicts eye movements in first- and second-language readers of diverse writing systems (Kimchi et al., 2025), though its authors test no text-level aggregate of the kind used here.

At the level of the sentence, structure imposes its own cost. Mean dependency distance indexes the distance over which a dependency has to be resolved, and clause density, subordination and noun-phrase depth index how many dependencies are open at once. That cost is not reducible to predictability. In 368 adults reading syntactically challenging sentences, language-model surprisal from 409 estimators approximated the early effects of disambiguation but not the additional rereading that repairing a structure incurred (Timkey et al., 2026), which is why the grammatical features form a family of their own and are never absorbed into surprisal. Gibson's (1998) theory separates the cost of keeping a prediction alive from the cost of integrating the element that completes it, and retrieval-based accounts place the cost at the point of resolution, where intervening and similar material slows retrieval (Lewis & Vasishth, 2005). Expectation-based accounts add that material before a head can make the head more predictable (Levy, 2008), so that in head-final configurations a longer dependency need not cost more. A clause-final verb has been read faster after longer intervening material in German and in Hindi (Konieczny, 2000; Vasishth & Lewis, 2006), and the strength of a locality effect is expected to vary with a language's statistics (Futrell et al., 2020). The grammatical slope is therefore allowed to vary by language in every model below.

On a capacity account, children, whose verbal working memory is still developing, should bear these costs disproportionately, and working-memory span does track complex-syntax comprehension and production between the ages of 5 and 12 (Delage & Frauenfelder, 2019). Whether the cost itself is proportionally larger in younger readers is a separate question, examined under Hypothesis 2. The evidence at the grain this study uses is less settled. In naturalistic corpus reading by adults, a distance-based integration cost predicted reading times at nouns and at auxiliaries but not at words in general, whereas word predictability predicted them throughout (Demberg & Keller, 2008). In the

Multilingual Eye-movement Corpus, global measures of syntactic and lexical complexity, among them the longest dependency path, the proportion of embedded sentences and mean sentence length, predicted adults' passage reading negligibly once passage length was accounted for (Kuperman et al., 2025). At the sentence level, only sentence length and position carried effects of any size. That study's lexical measures proved as negligible as the syntactic ones, but its authors separate such global measures from the aggregate properties of the individual words a passage contains. The lexical component here is anchored on the aggregates, mean frequency, word length and surprisal, whose word-level effects are well replicated across languages (Kuperman et al., 2024; Wilcox et al., 2023), and lexical diversity contributes to that component without being an anchor. Whether a text-level grammatical measure carries a reading-time cost for children reading whole stories therefore remains open. Hypothesis 1 is written so that a practically null grammatical coefficient would replicate that adult result in children, while a non-negligible one would qualify it.

At the level of the passage, what matters is whether the text states the connection between one idea and the next or leaves the reader to supply it. Sentence-to-sentence semantic similarity, overlap in referring expressions and the density of explicit connectives together index how much of that work the surface of the text has already done (Crossley et al., 2019). These are measures of cohesion. Coherence is a property of the representation the reader builds, and cohesion eases its construction without guaranteeing it (McNamara et al., 2010). Referential overlap descends from the rule that connects propositions sharing an argument (Kintsch & Dijk, 1978), which the construction-integration model retains in the weights it assigns to propositions close together in the text base (Kintsch, 1988). Semantic similarity between adjoining sentences approximates that propositional analysis (Foltz et al., 1998). An explicit relational marker speeds processing of the segment it introduces without changing what is recalled (Sanders & Noordman, 2000). Multi-dimensional characterisations of difficulty built from such indices outperform the classical readability formulae they replaced (Crossley et al., 2023). Evaluated against eye movements, readability scorers account for real-time reading ease only partially (Gruteke Klein et al., 2026), which is one reason this study models the psycholinguistic features themselves and never a composite readability score.

The systematic review that accompanies this protocol sharpens that general account in ways that shape the present design, and Supplementary Section S2 tabulates the studies closest to it in method. Text difficulty does not reduce to word frequency: in German fourth-graders, estimated lexical coverage predicted comprehension non-linearly (Ludewig et al., 2023). In 158 children aged 8 to 11, word frequency, pronoun density, concreteness and deep cohesion each explained unique variance in passage comprehension after decoding and oral language were taken into account, and lexical arousal explained variance beyond them (Pickren et al., 2022). That warrants carrying affective and discourse variables alongside conventional readability measures. Syntactic complexity has been shown to impair comprehension accuracy and slow recall in bilingual children (Wasito Kirana et al., 2025). In 5,133 second-graders reading informational text, word concreteness, referential cohesion and deep cohesion each predicted oral reading rate beyond the classical readability formulae (Tortorelli, 2020), and in first-graders a text-level vocabulary aggregate predicted rate, accuracy and comprehension (Bruner et al., 2026). Both samples are younger than this study's bands and both read aloud, so they establish that text properties reach a rate measure without fixing the size of that reach for silent reading at 7 and over.

Two further strands bear on the present design because they concern the reader and the reading

context. The first is the child's history of reading. Longitudinal work consistently shows that this history cannot be read as a simple dose of practice. Using random-intercept cross-lagged models from ages 7 to 16, Torppa et al. (2020) found that weaker early reading predicted less subsequent leisure reading, while later within-person increases in book reading predicted gains in comprehension. A ten-year study likewise found reciprocal associations between skill and print exposure, with the skill-to-practice path most visible early and accumulated exposure becoming more predictive later (van Bergen et al., 2021). Accumulated use of an app therefore carries both practice and the stable differences in skill and motivation that produced the use in the first place, so we will estimate within-child change and between-child differences separately. The setting matters as well as the quantity, since digital leisure reading and print reading do not relate to comprehension in the same way (Altamura et al., 2025), and digital leisure reading has its own association with comprehension independently of print (Altamura et al., 2026). What H6 observes is reading inside one app.

The second is narration. Read-aloud audio is often treated as a uniform benefit, and the evidence does not support that reading. A meta-analysis of 30 reading-while-listening studies, 62 effects over a total N of 1,945, found a small pooled comprehension advantage ($g = 0.18$). There was no reliable advantage where ordinary reading was self-paced ($g = 0.06$, 95 per cent CI [-0.07, 0.19]), and a larger one only where ordinary reading was paced by the experimenter ($g = 0.41$, 95 per cent CI [0.13, 0.70]) (Clinton-Lisell, 2023). Among students with reading difficulties, text-to-speech tools showed a moderate pooled effect ($d = 0.35$) that fell to $d = 0.15$ and included zero in within-participant designs (Wood et al., 2018). Neither synthesis settles the mechanism, which a large naturalistic corpus is well placed to address. If a spoken rendition works by supplying word identity without print-to-sound mapping, it should remove much of the cost that lexical difficulty imposes while leaving the cost of connecting ideas largely intact. If narrated reading time is instead set by the duration of the recording, the text properties keep only the purchase they have on the recording itself. A recording runs longer per word where words carry more syllables, so under pacing the lexical slope converges on the recording's own duration function, while the continuity slope, which no property of the recording tracks, converges on zero. Hypothesis 4 is written to separate these accounts.

The texts this platform serves are a population of stimuli produced by one process, and a process need not distribute text properties as human authorship does. Comparisons in other domains report that large-language-model output carries a distinct noun-heavy, informationally dense register even when prompted otherwise (Reinhart et al., 2025) and that human writing shows more scattered sentence-length distributions and more varied vocabulary than model output (Muñoz-Ortiz et al., 2024). In creative writing, the domain closest to the stories read here, stylometric comparison separates the two on measurable features of style (O'Sullivan, 2025). Whether the lexical, grammatical and discourse determinants established on human-written text describe reading across such a population is a question about the scope of that account (Haeffel & Cobb, 2022; Yarkoni, 2022), and the literature bearing on it can be counted.

On 26 August 2026, a Scopus search over 2018 to that date returned 3,701 records on children's reading comprehension, 6,592 on computational text features and 133,132 on platform log data. The pairwise intersections were 17, 28 and 45 records, and the three-way intersection was empty. That count sizes one database under one set of expressions. It did not return the log analysis of

this platform described below, published two months before the search, so the empty cell should be read as showing that the intersection is thin and unevenly indexed, not that it is unoccupied. The script that issues these searches and archives their counts is deposited with this protocol (Bernabeu, 2026c), Figure 1 shows the sets, and Supplementary Section S4 records the search and the surrounding publication trend. Over the same window, records combining children's reading with machine-generated text went from none before 2024 to two in 2024, six in 2025 and four in the first eight months of 2026. That small strand consists of classroom experiments that report gains in comprehension and motivation but describe the texts only qualitatively (Balantekin, 2026; Bulut & Aktepe, 2026). Beyond that count, generated and human-written texts have been compared directly, though not yet in children reading for themselves. Generated stories carried fewer personal pronouns and more positive emotion than 100 human-written ones, and adult readers were less transported by them (Appel et al., 2025). The models tested for children's stories could not hold their vocabulary to a level appropriate for younger readers (M. Valentini et al., 2023), which bears on the lexical component directly.

Let's Story, a feature of the *Applaydu* app developed by Gameloft and Ferrero International, combines the three topics of that count. Families choose a setting, a hero, a companion, a plot and a theme for each generated story. A generative model writes a narrative around those choices, and the app renders it as an illustrated storybook and, at the family's option, reads it aloud (Ratner et al., 2025). Every completed reading is logged. Comparable platforms have been studied one feature at a time. Booton et al. (2023) reviewed 11 studies and found inbuilt narration to aid comprehension and word learning. Log data from a research-oriented platform track ability, with reading speed in the BookBites app correlating with standardised scores at $r = .39$ to .48 (Sortkær et al., 2021). The commercial edutainment sector, whose user base is far larger, is described in the located literature mainly by content analyses and small experiments (Chatzopoulos et al., 2023; Furlong et al., 2025). One analysis of logs from *Let's Story* itself has now appeared. Over 419,700 story sessions in 18 languages across the thirty days following the feature's launch in November 2024, Ratner & Murphy (2026) describe how children of 4 to 9 meet challenge, reporting completion, where in a story a session was abandoned, and the post-reading difficulty rating. That study is descriptive by design and drawn from aggregated dashboard counts, so its authors state that it cannot follow an individual reader.

That study therefore leaves every association registered here untested. No reading time is modelled, and the session-level structure a within-child term requires was not in the summaries it drew on. Its window is the first month after launch, where the window analysed here opens more than a year later, so the novelty effects its authors name apply much less to this study. What it does establish is that the difficulty rating, the completion gate and the session record are analysable at scale on this platform.

A machine-checkable theory object deposited with this protocol holds the theoretical commitments behind the hypotheses. It names twelve constructs, eighteen propositions each carrying the mechanism it claims, six auxiliary assumptions and six rival accounts, and records the proposition each prediction derives from and the rival it is meant to separate. The object claims that the time a child spends per word is set by the costs a text imposes at the word, the sentence and the passage. Those costs are discounted by three reader capacities that these data do not measure and for which the age band stands in, and a spoken rendition bears on the first of them more than on the third. Five

further propositions carry the same constructs to the quality of the mental model a child ends with, which Figure 2 draws and which are tested only under the contingent hypotheses, if comprehension items prove to exist. Figure 2 is generated from that object. The object passes all twelve items of the versioned rigour checklist implemented in theoryforge, scoring 90.7 out of 100 (Supplementary Section S1) (Bernabeu, 2026d). Its causal graph is acyclic, so implications can be derived from it mechanically: that the three text properties are mutually independent given the reading level and the generation route, that narration is unrelated to accumulated reading, and that the difficulty rating adds nothing about reading level once reading time is known. We will test these implications before interpreting the confirmatory models, against the criterion registered under Outcome-neutral quality checks, and we will report any failure.

This study is observational throughout. Children choose their stories, their reading level and whether to switch narration on, and families choose whether to keep using the app. A laboratory experiment establishes that a text property *can* move reading when an experimenter controls the text, the setting and the task, not that it *does* when nobody controls anything. That second question is one about generalisation, and it cannot be settled in the setting that produced the original result (Nastase et al., 2020; Yarkoni, 2022). Whether a relation holds in a new population of texts, languages and readers is in that sense a test of the account that predicts it (Haeffel & Cobb, 2022). In the logs of a vocabulary-learning app, an effect of valence established under laboratory control held, while an effect of concreteness did not (Wild & Kuperman, 2024), though those readers were adults and the outcome was learning. A laboratory result need not survive the move to platform data, and Hypothesis 1 is written so that it need not.

The platform's first advantage is stimulus sampling. Reading research has known since Clark (1973) that treating a handful of experimenter-written texts as a fixed set inflates confidence in conclusions meant to hold for texts in general, and the mixed-model solution has been available for over a decade (Judd et al., 2012). The design here treats Text as a crossed random factor over hundreds of texts that children selected for themselves, which is both the prescribed remedy and, in Brunswik's sense, a representative design (Dhami et al., 2004). A varying intercept does not by itself make an inference general where the variation between texts is systematic (Bear & Phillips, 2022), so the systematic sources the platform records are carried by the model itself. Because an average taken over texts can also hide the variation between them (Simonsohn et al., 2025), the text-level deviations will be displayed against the component scores at Stage 2. We avoid the phrase *ecological validity* throughout. It is used in at least two distinct senses, Brunswik's and Orne's (Kihlstrom, 2021), and has been argued to be too ill-defined to bear on generalisation at all (Holleman et al., 2020). We make the narrower, checkable claim about how the stimuli were sampled.

Linguistic breadth is the second gain. English dominates cognitive science to a degree that warps which mechanisms are studied and licenses overgeneralisation from one atypical language to the species (Blasi et al., 2022). Reading is among the worst affected, because writing systems differ in ways that matter for every stage of processing. The most ambitious response to date, the Multilingual Eye-movement Corpus, assembled eye-tracking during text reading in 13 languages, since extended to 21 (Siegelman et al., 2022; Siegelman et al., 2025), and its authors report that the existing empirical base is insufficient for cross-linguistic comparison. A commercial platform reaches many of the same languages incidentally, in far larger numbers of readers per language, and reaches children where the corpus records skilled adults. MECO records eye movements while this study

records only whole-story duration, so the platform gains breadth and loses resolution. The direction of each of the three associations is expected in every one of these ten languages. The operations reading recruits, mapping a written form to sound and to meaning and relating it to what came before, are common to writing systems that differ in almost everything else (Frost, 2012; Verhoeven & Perfetti, 2022), and the adult evidence bears this out for word length and frequency across twelve alphabetic languages (Kuperman et al., 2024). Magnitude is another matter. The lexical association is expected to vary with how a script encodes the word. The grammatical association is expected to vary with a language's word order and statistics. No comparable reason has been derived for referential continuity, which operates above the word. The language-varying component slopes are where these expectations are visible, they are reported for every language, and no verdict in Table 1 rests on any one of them.

The third gain is behaviour as it occurred. Logged and self-reported digital media use diverge systematically enough to have warranted a meta-analysis of the discrepancy (Parry et al., 2021), and in 109 French primary-school children parents' estimates of weekly reading time were three times the logged figure, which predicted fluency better (Brossette et al., 2026).

Against this stands the strongest statistical objection to any large non-random dataset. Meng's (2018) decomposition shows that the error of a sample mean is the product of three quantities: data quality, data quantity and the difficulty of the problem. The first is the correlation between the recorded value and whether it was recorded at all, and probability sampling holds it near zero. Once it is not held there, the population size stops cancelling: for any given sample size, estimation error then grows with the square root of the population. Bradley and colleagues (2021) gave the canonical demonstration: two surveys collecting roughly 250,000 responses a week and 75,000 a fortnight overestimated United States COVID-19 vaccine uptake by 17 and 14 percentage points, with margins of error so small that the wrong answers looked precise, while a well-designed panel of about 1,000 responses a week was accurate. Every selection gate in this study is a candidate for that correlation: who owns a tablet, who installs an app tied to a confectionery brand, who registers consent for research use, who switches narration off, and who finishes a story.

That objection is real, and it also has a boundary. Meng's result concerns the estimation of a population mean, and this study estimates no population mean. Its estimands are conditional associations inside the sample, within a language and within a reading level. A gate that operates on the reader alone, such as who owns a tablet or who registers consent, changes which children a slope describes while leaving the slope itself intact. Where the reader attribute also moderates the slope, the estimate becomes the average slope across the families who use the platform, which is how an estimate of a heterogeneous association from a non-random sample should be read (Bryan et al., 2021). A slope is distorted only when the gate is a common effect of the text property, or of something that moves with it, and of the outcome or of an unmeasured cause of the outcome, a narrower condition than the one that wrecks a mean (Elwert & Winship, 2014; Hernán et al., 2004; Lamp & MacKinnon, 2025). Of the gates listed above, only completion meets it, and it is treated next. Narration is a state the models condition on, so the selection it carries falls on the denominators of Hypothesis 4, where it is treated. We will report these coefficients as estimates for children who read on this platform, and we will not present them as population values for children in general, nor will we report a population mean of reading speed for any group.

The extract contains completed readings only, so the analysis conditions on completion, which

is plausibly a common effect of the text's difficulty and the reader's capacity. Conditioning on a common effect can induce an association between its causes where none existed (Rohrer, 2018). The direction here is predictable: if harder texts are abandoned more often, the hard texts that survive into the data are disproportionately those read by children who could manage them, which attenuates the very coefficients Hypothesis 1 predicts. If younger children abandon demanding texts more often, their surviving readings of such texts are the more selected and their slopes the more attenuated, which on the sign convention above pushes the grammatical interaction positive, towards the bottleneck-release account, and the continuity interaction negative, towards the cue-utilisation account (Elwert & Winship, 2014; Hernán & Monge, 2023). We will therefore report the completion rate by age band, where the data controller can supply it, beside any determinate verdict on Hypothesis 2 or 3. Where the younger band's rate is the lower, a verdict supporting the bottleneck-release or cue-utilisation account will be reported as consistent also with differential attrition. The bias therefore works against our own prediction, so a non-negligible result survives the objection more comfortably than a null does, and an inside-region verdict for Hypotheses 1 to 3 must be reported with the completion rate beside it.

Abandoning a text is not the only way of escaping it. A reader can also meet a demanding text by reading it less carefully. Where reading speed was manipulated, comprehension held across a broad band of rates and fell only at the fastest, around 405 words per minute, while the influence of word properties on eye movements persisted throughout (Schwalm et al., 2026). That result cuts two ways here. It places the plausibility bound of 400 words per minute close to the rate at which comprehension did give way, so the bound removes page-flipping while leaving fast but still comprehending reading inside the sample. It also gives no warrant for assuming that the faster readings carry weaker text effects, since the word-level effects in that study did not weaken with speed. Neither the extract nor the design can separate a skimmed reading from a fluent one, so we register the ambiguity rather than a direction for it.

Nothing here is randomised, and we do not use causal verbs for variables that families chose. That restraint is not the same as pretending the questions are not causal: euphemism has been argued to damage observational research by obscuring what is actually being asked (Grosz et al., 2020; Hernán, 2018). We state the causal structure explicitly in a deposited graph, derive the adjustment set from it, register the rival explanations it implies, and then report associations conditional on that adjustment.

Secondary data also invite the analytic flexibility that preregistration exists to remove, and the field has standards for preregistering analyses of existing data (Van den Akker et al., 2021; Weston et al., 2019). The arrangement described under Data provenance goes further. The extract is absent and unanalysed. The data controller holds it, and it reaches us only after in-principle acceptance, so the most common objection to observational work, that the analyst saw the data before choosing the analysis, does not arise. Holding the outcome out of reach until the analysis is fixed is the principle of blind analysis (Dutilh et al., 2021; MacCoun & Perlmutter, 2015).

This study does not attempt to predict a child's reading. It estimates a handful of pre-specified associations whose attainable precision we calculated before the data were released, and says in advance which of them the available records cannot settle. Group-level associations need not describe any individual, and variance around an expected value was two to four times larger within people than within groups (Fisher et al., 2018). Tier 2 therefore separates each child's own change

from the differences between children, and the two are never reported as a single quantity.

Three consequences of all this are built into the design. First, the reading level a family sets determines which version of a text the platform serves, so text difficulty is partly assigned by a choice that also reflects the child's capacity. We will therefore estimate the text coefficients within reading level and check whether they survive that restriction. Second, the app records elapsed time for a whole story and never for a page, so the outcome is a rate, and we state below which analyses that forecloses. Third, the data carry no measure of any individual child's reading skill, vocabulary or language background, so age band stands in for a bundle of developing capacities and is interpreted as nothing finer. The reading language a family selects need not be the child's first language, so every coefficient below averages over children reading in a first and in a second language. In adult bilinguals reading paragraphs, frequency effects were larger in the second language than in the first and grew as second-language exposure fell (Whitford & Titone, 2012). A reading language is therefore not a language background, and the country-by-language analysis registered under Sensitivity analyses is the only purchase the extract gives on this.

Under the simple view, reading comprehension is the product of decoding and language comprehension, and decoding's share of the variance falls across the primary grades (Gough & Tunmer, 1986; Hoover & Gough, 1990; Lonigan et al., 2018). Automaticity theory says why. Identification that runs without deliberate attention releases capacity for the work of integration (LaBerge & Samuels, 1974; Perfetti, 2007), and the reading systems framework separates word identification, word-to-text integration and the situation model (Perfetti & Stafura, 2014), which is the partition the three components are built to instantiate. Phonological recoding on each new word is the mechanism by which reading itself produces that automaticity (Share, 1995), which ties Hypothesis 6 to the same account. In the eye-movement record, reading efficiency rose by 36 per cent between second and fourth grade in a longitudinal sample, where linguistic skill determined the change and oculomotor skill did not (Huestegge et al., 2009). In 194 Greek-speaking children, silent reading speed rose from 111 words per minute in grade 3 to 166 in grade 6, through fewer and shorter fixations and longer saccades, and those oculomotor parameters explained variance in speed beyond vocabulary and cognitive measures (Gleni et al., 2026). The age band is therefore read as a coarse proxy for reading experience and the automaticity it confers, in the aggregate and never for one capacity.

The boundary is not ours to set, since the two age bands available to us fall at 7 to 8 and 9 or older. The platform describes its audience as children of 4 to 9 (Ratner & Murphy, 2026), so the older band is expected to sit mostly at its lower edge, and its upper edge is not recorded. Whether change across such a boundary is governed by attained reading level or by the maturation of oculomotor control is still argued (Justino & Kolinsky, 2023; Louwen et al., 2026), and the finer-age sensitivity analysis registered below is what these data allow on the question. The LexicalDifficulty × AgeBand interaction is estimated in the tier-1 model and reported, but is not among the confirmatory tests, because the eye-movement literature establishes larger absolute lexical effects in children (Tiffin-Richards & Schroeder, 2015) without settling whether they are larger as a proportion of a slower baseline, the quantity a log outcome measures.

The quantity these hypotheses share is the time a child spends per word over a whole story. In the laboratory, reading time carries information about processing because a word is looked at for about as long as it is being worked on (Just & Carpenter, 1980). The properties that lengthen fixations also predict passage-level eye-movement measures on texts of more than two hundred words, shown

so far in second-language readers of English (Nahatame & Yamaguchi, 2025). The theory object records that step as the auxiliary assumption that elapsed time indexes effort, on which six of its ten predictions depend. The positive control on the timing column registered under Outcome-neutral quality checks puts it at risk. A whole-text rate is also more than the sum of its word-level costs, since the efficiency of processing items in a series explained variance in reading connected text once discrete word reading was controlled (Romero et al., 2024). The components in this design are properties of the text, so reader-side efficiency sits in the age band and in the residual. We describe the outcome as reading pace, an index of the processing demand a text imposes.

One rival reading of that same interval is registered in the theory object. It traces time on a story to how engaging the story is, where this study traces it to how demanding the text is. Among 122 sixth-graders reading silently, the difficulty of the passage, rated situational interest and their interaction all predicted time on task, and interest accounted for the most variance (Welland & Burns, 2025). Two registered predictions are marked diagnostic against it. The first is a continuity coefficient non-negligible in the positive direction, which is what an engagement reading would produce, since a text that states more of its connections might hold a child longer where the theory predicts it should release them sooner. The second is Hypothesis 5: a difficulty rating formed from effort the child noticed expending should rank the three components as reading time does, where a rating tracking enjoyment need not. Neither prediction disposes of the rival on its own.

Six confirmatory hypotheses follow. Two of them, the interactions between the text properties and the age band, are registered as two-sided tests that adjudicate between rival accounts, because the direct developmental evidence in this age range supports neither sign. LexicalDifficulty and GrammaticalLoad are oriented so that higher scores mean a more demanding text, and Referential-Continuity so that higher scores mean a text that states more of the links between its ideas. Signs are fixed before any component score is joined to an outcome. The age band and narration enter the models as deviations from their observed prevalence: a reading scores 1 for the older band, or for narration on, and 0 otherwise, minus the proportion of tier-1 readings in that state. Each component’s main effect is then its prevalence-weighted average slope across the two states, which carries no design effect from imbalance, and each interaction is the difference in slope between the two states, positive when the slope is larger in the older band or under narration. The outcome for Hypotheses 1 to 4 and 6 is log reading time per word, standardised within language, so a coefficient is a within-language standardised slope and the region of practical equivalence is 0.05 standard deviations. Hypothesis 4 concerns a difference between two attenuation proportions, with a region of 0.05 on that difference. Hypothesis 5 uses the child’s own three-point difficulty rating on an ordinal scale, where the region is a half-width of log(1.05) on the log-odds scale, an odds ratio between 0.952 and 1.05.

Because the two main effects run in opposite directions, the same developmental claim takes opposite signs in the two interaction terms. Grammatical load is associated with slower reading, so a weaker association in older children makes a positive slope smaller and the interaction negative. Referential continuity is associated with faster reading, so a weaker association in older children makes a negative slope less negative and the interaction positive.

The outcome is the logarithm of reading time. Children in the two age bands differ in overall reading rate, and a difference in rate that is purely multiplicative manufactures a spurious interaction on a raw time scale (Faust et al., 1999). On the log scale, a multiplicative difference in rate appears

as a main effect and never as an interaction, so every interaction reported here is a difference in *proportional* cost.

Two things the log scale does not fix are registered here in advance. The first is that the transformation does not remove an interaction produced by an additive component of the interval. If the time a word costs is a base decoding time plus a grammatical increment, the log-scale grammatical slope is that increment divided by the base. A reader with a smaller base then shows a larger proportional slope even where the increment is identical. The automaticity account asserts an additive decoding component of just this kind, so a positive Hypothesis 2 interaction is that account's expected signature on this scale, and the transformation does not manufacture it. The second is that part of any whole-story interval is spent on the illustrations, on turning pages and away from the task. Children's attention to the text of an illustrated book rises across the primary grades (Roy-Charland et al., 2007), so that non-reading share may be larger in the younger band, making that band's proportional slopes smaller. That moves the Hypothesis 2 interaction towards the positive sign and the Hypothesis 3 interaction towards the negative one. These data cannot separate a smaller base interval that is faster decoding from one that is less time away from the text, so a determinate verdict of either sign will be reported with that limit stated beside it.

The outcome is also divided by word count, and that choice has a rationale and a residue of its own. In the standard cohesion manipulation, the more cohesive version of a text is 20 to 40 per cent longer, so a duration aggregated over a whole text would show higher cohesion slowing reading for arithmetic reasons alone. Dividing by word count and retaining log text length as a covariate closes that path, and a raw duration is never used as an outcome. Dividing by words does not divide out sentence boundaries, at which readers spend extra time whose cause is unsettled (Rayner et al., 2000; Warren et al., 2009), so a text of shorter sentences carries more boundaries per word. That works against the expected sign of sentence length taken alone.

Table 1 sets out each hypothesis against the sampling and analysis plan that tests it, the precision the design analysis says that test requires, how each possible outcome will be read, and what would be shown wrong. We fixed every row before any outcome data existed.

Table 1

Study design table. Eligibility for every row: sessions the platform flags as incomplete, crashed or duplicated are removed, as are accounts flagged as duplicates under the platform's own rules. A story needs a positive word count, and a reading faster than 400 or slower than 10 words per minute is excluded. Only an account's first reading of a text enters tier 1, and readings sharing a session-timeout value are treated as right-censored where the registered rule triggers. No datum is ever replaced.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Are the text properties that govern reading in the laboratory associated with reading pace in an app?	H1. LexicalDifficulty and GrammaticalLoad are positively, and ReferentialContinuity negatively, associated with log reading time per word.	Tier 1: every eligible completed reading in the frozen extract, one row per child and text, children recorded in age band 7-8 or 9+, under the eligibility rules in the table note. There is no stopping rule, because the extract is fixed before modelling. ROPE of 0.05 *SD*. A determinate verdict on an effect of 0.10 *SD* with probability 0.90 needs 738 distinct texts for the lexical coefficient (95 per cent interval 640 to 843), 660 for grammatical load (593 to 731) and 634 for continuity (568 to 704), and at 0.95 they need 926, 812 and 784. A threshold will be re-derived if the review's pooled estimate for that component falls below 0.10 *SD*, as registered under Design and precision analysis. Below its threshold a coefficient is declared underpowered before fitting.	The tier-1 Bayesian model specified under Confirmatory models: within-language standardised log time per word on the three components, the age band, narration and their interactions, adjusted for reading level, text length, valence, arousal, narrativity and provenance, with varying intercepts for Text, Language, Country and App version, varying component slopes by Language, and an Account intercept where the estimability rule is met. The three component posteriors are the registered tests.	Each coefficient is labelled non-negligible in the predicted direction, non-negligible in the opposite direction, practically equivalent to zero, or inconclusive. A combined H1 verdict is recorded only when all three are non-negligible in the predicted direction. A coefficient non-negligible in the opposite direction counts against the corresponding proposition of the theory object. A positive continuity coefficient is not read as a reverse cohesion effect, which is a learning outcome whose mechanism predicts slower reading of the less cohesive text. Equivalence is evidence that the property's laboratory effect does not reach 0.05 *SD* in this setting. Where the compression diagnostic fires, the derived unnarrated slope carries a component's verdict if the unnarrated readings cover its threshold; otherwise the pooled verdict stands, reported beside the unnarrated slope and as consistent with the recording having set the pace. Shown wrong if all three are equivalent: that the lexical, grammatical and discourse determinants of reading established under laboratory control operate on unsupervised reading of machine-generated stories.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is the association between grammatical load and pace proportionally smaller, or larger, in the older band?	H2. GrammaticalLoad × AgeBand, tested two-sided. The capacity and experience accounts predict a negative interaction, the bottleneck-release account a positive one.	As H1. Tier 1 admits only readings with a recorded age band. A determinate verdict on an interaction of 0.10 *SD* needs 1,230 distinct texts (1,107 to 1,359) at an average of twelve readings each for 0.90 and 1,510 (1,349 to 1,680) for 0.95, and declaring equivalence needs about 1,600. Both figures assume a balanced band split and will be multiplied by $1/(4p(1-p))$ at the observed band prevalence before fitting.	The same tier-1 model. The GrammaticalLoad × AgeBand posterior is the registered test. Cross-validated comparison against a model without the interaction will be reported alongside and does not gate the coefficient.	Negative and outside the ROPE supports the capacity and experience accounts, which it does not separate, and positive and outside supports the bottleneck-release account. Inside the ROPE is evidence that no developmental signature reaches 0.05 *SD* across these two bands, with the caveat that a non-monotonic trajectory cancelling across two coarse bands would produce the same posterior, as would interactions of opposite sign in orthographies of different depth. The per-language interactions are reported beside the pooled one, and the finer-age sensitivity analysis can display a reversal where the field permits, being fitted in an unordered as well as a monotonic coding. Shown wrong: whichever account or accounts the sign contradicts. A posterior inside the ROPE weighs against all three, since each predicts a signed proportional change.
Is explicit connection between ideas associated with a proportionally larger gain in pace for the younger or the older reader?	H3. ReferentialContinuity × AgeBand, tested two-sided. The knowledge account predicts a positive interaction, the cue-utilisation account a negative one.	As H2. A determinate verdict needs 1,431 distinct texts (1,276 to 1,596) for 0.90, and the 0.95 target was not reached within the 1,600 texts simulated. The probability of correctly declaring equivalence approaches 0.90 at 1,600 texts. The band-prevalence adjustment applies as for H2.	The same tier-1 model. The ReferentialContinuity × AgeBand posterior is the registered test. Connectives enter the component by relation type, with causal and contrastive connectives as anchors.	Positive and outside the ROPE supports the knowledge account, on which continuity substitutes for an inference the younger reader finds costly. Negative and outside supports the cue-utilisation account, on which a connective must first be reached and understood before it can help, and is also consistent with a lower standard of coherence in the younger band, which this design does not separate from it. Inside the ROPE is evidence for neither, with the same non-monotonicity caveat as H2, and with the further caveat that anchors carrying opposite-signed interactions can cancel within the component, which the registered per-anchor check reports. If the retained component loads below .30 on both connective densities, the sign is read for referential and semantic continuity only and the adjudication between the two accounts is reported as not addressed. Shown wrong: whichever account the sign contradicts.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is narration associated with a proportionally smaller lexical coefficient than the continuity coefficient?	H4. The proportional attenuation of the lexical coefficient under narration exceeds that of the continuity coefficient by more than 0.20.	As H1, evaluated only where both narration states occur within the eligible sample. The within-account contrast becomes the primary estimate when the accounts observed in both states cover at least the adjusted text requirement. ROPE of 0.05 on the difference of proportions. The simulated figure of 1,234 texts (1,113 to 1,361; 1,506 at 0.95) prices the lexical-by-narration interaction coefficient and is a lower bound for the difference rule. It will be multiplied by $1/(4p(1-p))$ at the observed narration prevalence, about 3.0 at an unnarrated share of 0.09, and stated before fitting.	Draw-by-draw contrast of the two proportional attenuations, computed from the derived narrated and unnarrated slopes of the tier-1 model under the rule given under Confirmatory models. Gated on the derived unnarrated slopes of the lexical and continuity components both being practically non-negligible. Reported with the residual standard-deviation ratio and repeated in the narrated sessions the audio cannot have paced.	At least 95 per cent of the posterior of the difference above 0.20 supports selective removal of decoding cost. More than 95 per cent inside the region supports the account on which the recording sets the pace, with the duration-adjusted contrast reported beside it. More than 95 per cent below -0.05 is consistent either with reverse selectivity, which contradicts both accounts, or with pacing acting through the recording's duration, which the duration-adjusted contrast reported beside it separates. Anything else is inconclusive and both accounts remain live. If the gate fails, H4 is reported as not evaluable. If the compression diagnostic fires, the full-sample verdict is withheld and H4 rests on the unpaced subset. Shown wrong: either the decoding-scaffolding account of narration or the audio-pacing account. The contrast is constructed so that they cannot both survive as the account of the same readers, and a determinate verdict reports which regime dominates the narrated sessions.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Do children's own difficulty judgements track the same text properties?	H5. LexicalDifficulty and GrammaticalLoad are associated with higher odds, and ReferentialContinuity with lower odds, of a harder rating, each outside the region, and the three rank by magnitude as in the reading-time model.	Tier-1 sessions carrying a rating. A text rated by fewer than five distinct children does not contribute. The primary model is complete-case, with an inverse-probability sensitivity analysis for non-response. ROPE of a half-width of log(1.05) on the log-odds scale. At the 30 per cent availability the structural sample showed, the 0.90 target is not reached within the 1,600 texts simulated and projects to roughly 9,300 to 10,300 generated texts, about 2,700 to 3,000 carrying at least five ratings (Supplementary Section S5). The simulation will be re-run at the observed availability before fitting. This is the hypothesis most likely to be declared underpowered.	Ordinal cumulative-logit model with the same fixed and grouping structure and language-varying thresholds. The ordering claim is the posterior probability that the three coefficients rank by magnitude as they do for reading time, read against the 0.90 criterion registered under Decision rules.	All three odds ratios outside the region in the predicted directions, with ordering probability at least 0.90: two readouts of the same expended effort, reached by different routes, converge on the same ordering of text demands. Predicted directions but ordering probability below 0.90: partial convergence, with the rating usable as a difficulty signal but not as a rank measure. Odds ratios inside the region while the reading-time coefficients are non-negligible: the rating is not a usable measure of experienced difficulty, and the ordering is not interpreted. If the older band's odds ratio then lies outside the region, the limit is reported as specific to the younger band and the pooled verdict stands. Both outcomes null: the components themselves are suspect. If underpowered, the ordering is reported descriptively and convergent validity is recorded as not established. Shown wrong in the third case: that reading time and reported difficulty read out the same expended effort. Shown wrong in the second: that a child's judgement weights the three sources of demand as reading time does.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Within a child, is more accumulated in-app reading associated with a faster pace?	H6. Within a child, accumulated in-app reading, entered on a log scale and standardised, is negatively associated with log time per word.	Tier 2: children with at least three eligible sessions on distinct texts. The simulated requirement is 609 children (528 to 697), each reading five distinct texts, for 0.90 and 731 (621 to 851) for 0.95, for the unadjusted trend. The tier-2 sweep will be re-run at the observed sessions-per-child distribution with the registered covariates added, and the hypothesis is declared underpowered, not untestable, below the re-run threshold.	The tier-1 specification refitted on tier 2 with a child varying intercept, decomposing accumulated words into the child's mean and the session's deviation from it, alongside session order, days since the previous session and days since the child's first eligible session. The deviation term is the registered test.	Negative and outside the ROPE: an association over the observation window, never a practice effect, because skill and practice are reciprocal over longer spans and maturation is the named residual alternative. Positive and outside: contradicts the orthographic-learning account and is read as within-window slowing that the covariates do not capture. Inside the ROPE: evidence against a within-window orthographic-learning signal of at least 0.05 *SD*. The account-mean coefficient is reported beside the deviation term and is not a registered test, because both accounts predict it negative. The difference between the two coefficients is reported as a contextual effect and read as consistent with, and not as evidence for, between-child selection, since the composition of accounts by age band, language and entry date, and the truncation of accumulated reading at the window's opening, also contribute to it. Shown wrong: that reading in the app sharpens the word-recognition machinery reading depends on, within an observation window over which the reverse path is checked by the registered attrition and lagged-pace refits.

Hypothesis 1: text properties and reading pace. We predict that, across distinct texts and within reading level and language, LexicalDifficulty and GrammaticalLoad will each show a positive association with log reading time per word, and ReferentialContinuity a negative one. The three coefficients are evaluated and labelled separately, and the failure of one does not suppress reporting of the others. An overall verdict on Hypothesis 1 is recorded only when all three are practically non-negligible in the predicted direction. A coefficient that is practically non-negligible in the opposite direction is read as evidence against the corresponding text-to-effort proposition of the theory object. A positive continuity coefficient in particular is not labelled a reverse cohesion effect. That effect is a learning outcome (McNamara et al., 1996; McNamara & Kintsch, 1996; O'Reilly & McNamara, 2007). On the construction-integration account it arises because a less cohesive text makes the knowledgeable reader infer more, which on a time scale would appear as slower reading of that text. A coefficient practically equivalent to zero is evidence that the property's laboratory effect does not reach 0.05 standard deviations in this setting, which for the grammatical component would extend to children the negligible passage-level effect of global complexity measures reported for adults (Kuperman et al., 2025). That reading does not carry over to the lexical component, whose anchors are word-level aggregates the adult study did not test at passage level.

One further rule follows from how narration is distributed. Narration was on in about nine sessions in ten of the structural sample, so each prevalence-weighted main effect is dominated by the narrated slope. If the recording sets the pace of narrated sessions, the continuity slope under narration converges on zero and the lexical slope on the recording's own duration function, as set out for Hypothesis 4 below. Where the compression diagnostic registered under Confirmatory models fires, the derived unnarrated slope therefore carries the Hypothesis 1 verdict for a component whenever the unnarrated readings cover at least that component's threshold in Table 1, with the prevalence-weighted slope reported beside it. Where they do not, the prevalence-weighted verdict stands and is reported beside the unnarrated slope, and an inside-region continuity verdict or a non-negligible lexical verdict is then reported as consistent with the recording having set the pace. The rule is fixed here and does not depend on the sign of any coefficient.

Hypothesis 2: grammatical load and the age band. We register the GrammaticalLoad × AgeBand interaction as a two-sided adjudicating test, because accounts of what changes between these ages divide over the sign on the scale this study uses. On the capacity account, a grammatical dependency costs time in proportion to how close its maintenance comes to exhausting the working memory available, so as that capacity matures the same structure costs proportionally less and the interaction is negative (Just & Carpenter, 1992; King & Just, 1991). The experience-based alternative to capacity shares that sign. On it what grows with age is exposure to complex structure, and exposure makes the structure cheaper (MacDonald & Christiansen, 2002). Children with more print exposure produce relative clauses more as adults do (Montag & MacDonald, 2015), which is evidence about production rather than about the cost of reading such a structure. The account that predicts the opposite sign holds that what changes is the base the proportion is taken over, and how fully structure is built on it. The time a word costs carries a decoding component alongside the cost of grammatical load, and as print-to-sound mapping automatises that component shrinks (LaBerge & Samuels, 1974), leaving the reader with representations whose quality makes retrieval cheap (Perfetti, 2007), so an unchanged absolute grammatical cost occupies a larger proportion of a smaller base and the interaction is positive. Comprehenders often leave syntactic structure shallow or unresolved, structure left unresolved is read faster, and the evidence for that claim is mixed (Ferreira et al., 2002;

Frances, 2024; Swets et al., 2008). Children of 8 to 10 show the same signatures as adults, and the older children among them attend more to the punctuation that marks clause boundaries (Wonnacott et al., 2016). A younger reader who builds shallower structure therefore incurs less of a dependency's cost. The capacity account is the theory object's own proposition, and the bottleneck-release account is registered there as its rival, so a positive interaction would count against that proposition as well as for the rival.

The capacity account's premise is less secure in this age range than its formulation suggests. Across 197 studies, the relation between working memory and reading was moderate, $r = .29$, and its contribution to comprehension was accounted for by decoding and vocabulary where those were controlled (Peng et al., 2018). In children, memory span has been related to relative-clause comprehension, though narrowly in both studies located: in German-speaking 7-year-olds the relation was to digit span and held for object relatives alone (Arosio et al., 2012), and in French- and German-speaking children verbal working memory correlated with object-relative comprehension only in French (Solaimani et al., 2025). The premise may therefore not hold uniformly across the languages in this sample, which is one reason this interaction is also reported separately by language.

The direct evidence does not decide between the accounts. The largest eye-movement sample over this band, 81 children reading 12 items, found no attenuation (Wonnacott et al., 2016). It manipulated temporary ambiguity, which indexes reanalysis, a different demand from the sustained maintenance that dependency length imposes, so its null does not transfer to the present operationalisation. Where an interaction has been reported, it was confined to late measures with fourteen children per group (H. S. S. L. Joseph & Liversedge, 2013). Recomputed from that study's tables as a proportion of baseline, the two measures disagree. On go-past time in the target region, our computation gives 15 per cent in the younger children, 25 per cent in the older and 31 per cent in adults, which is the pattern a positive interaction would reproduce on a proportional scale. On post-target total reading time, where the study's own significant interaction lay, the proportional effect ran the other way, from 27 per cent in the younger children to 12 per cent in the older and 4.5 per cent in adults.

A third possibility is that development over this band is not monotonic at all. The largest of the existing tests, with 250 children aged 8 to 11 reading and hearing sentences word by word, reported a U-shaped pattern in the online processing of restrictive relative clauses over that range (Booth et al., 2000). Offline comprehension of complex syntax grows most steeply between 5;6 and 7 years (Frizelle et al., 2025), so the younger band sits immediately after the steepest phase. The boundary between the bands also coincides with a documented shift in online processing. In Russian-speaking children, accuracy on passive sentences rose sharply between 7 and 8, and although 8- to 10-year-olds then matched adults on accuracy, the incremental-revision signature at the critical word, present in adults and at 8 to 10, was absent at 6 to 7 (Kruchinina et al., 2026). Accuracy converged before processing did, which is one reason a time-based outcome is the right grain for this interaction.

The three outcomes are read as follows. A negative interaction outside the region of practical equivalence supports the maturational accounts, capacity and experience, which it does not separate. A positive one supports the bottleneck-release account, and a posterior lying inside the region is evidence that no developmental signature reaches a detectable size across these two bands. A non-monotonic trajectory cancelling across two coarse bands would produce that same inside-region posterior, as would interactions of opposite sign in orthographies of different depth. The per-language interactions and the finer-age sensitivity analysis registered below are therefore reported beside the

pooled coefficient.

The two interactions are coupled under bottleneck release. The same dilution that raises the proportional grammatical cost in the older band deepens the proportional continuity saving. Bottleneck release therefore predicts a positive Hypothesis 2 interaction, a negative Hypothesis 3 one and a lexical-by-age interaction that is negative or null. The tier-1 model already estimates all three, and we will report the joint pattern descriptively beside the two registered coefficients. It gates no verdict.

Hypothesis 3: referential continuity and the age band. We register the ReferentialContinuity × AgeBand interaction the same way, because the construction-integration account of comprehension (Kintsch, 1988), set beside what exploiting a cohesive cue requires, supplies two moderators of coherence facilitation, both rising between 7 and 11 and pushing in opposite directions. On the knowledge account, a gap in the text is bridged from knowledge that the younger reader has less of and retrieves less readily. Age differences in coherence inferencing between 6 and 15 survived making the relevant knowledge equally available to every child, and its accessibility mattered more for the younger children (Barnes et al., 1996). At 11 to 12, higher knowledge went with faster reading of sentences that required an inference (Cain et al., 2024). Surface continuity substitutes for an inference that is costlier at 7 to 8 than at 9 and over, so it is worth most to the younger reader and the interaction is positive.

On the cue-utilisation account, exploiting a connective requires enough decoding fluency to reach it and enough explicit knowledge of what it signals, so the older reader gains more and the interaction is negative. Knowledge of causal and adversative conjunctions is still developing between 8 and 10 (Cain et al., 2005; Cain & Nash, 2011), and comprehension of connectives grows across the primary years (Volodina & Weinert, 2020). The cue-utilisation account runs through decoding fluency, so it implies a larger negative interaction where decoding is still maturing across the two bands, which the paragraph on orthographic depth places in English, French and Portuguese. The knowledge account, which runs through background knowledge and narrative experience, implies no such dependence. The per-language interactions registered under Sensitivity analyses are reported for that reason as well, and they are descriptive, gating no verdict on Hypothesis 3.

Decoding fluency and knowledge of the cue both develop well past the ages studied here. Knowledge of what a connective signals is still incomplete in French-speaking teenagers of 12 to 19 and in native Russian-speaking teenagers of 11 to 17, and in both samples vocabulary size and exposure to print were stronger correlates of appropriate use than age was (Tskhovrebova et al., 2022, 2023). Both moderators, background knowledge and knowledge of the cue, are themselves likely to grow with the reading a child accumulates, which is a further reason the net sign of the interaction is not derivable in advance. The knowledge account is likewise the theory object's own proposition, and cue utilisation is registered there as its rival, so a negative interaction counts against that proposition.

The inference the knowledge account has continuity sparing is the one that referential overlap and semantic similarity between adjacent sentences make unnecessary, whereas the cue-utilisation account concerns a connective. Semantic and causal relatedness dissociate by reader skill in adults, semantic relatedness weighing more for the less skilled reader (Todaro et al., 2010), and the two bear differently on processing time and on what is remembered (Wolfe et al., 2005). The three anchors of ReferentialContinuity need not therefore carry the same age interaction, and a component interaction

inside the region could arise from anchors whose interactions differ in sign. The per-anchor check registered under Sensitivity analyses does not gate the verdict. Where the component posterior lies inside the region while the anchor interactions lie outside it with opposite signs, the equivalence verdict stands and is reported with that pattern beside it.

A negative interaction has a second reading. On the Landscape model, whether a reader attempts a bridging inference depends on the standard of coherence they bring to the text (van den Broek et al., 2005; van den Broek & Helder, 2017). A reader who does not attempt it pays no time for the gap, so continuity would save the younger reader nothing if that band read to a lower standard. The online evidence bounds that concern. Children of 8 to 13 made the local inferences a coherent representation required, if not the optional ones (H. Joseph et al., 2021). Among the accounts on which continuity is worth more to the older reader, cue utilisation and a lower standard of coherence are not separated by this design, and a negative verdict is reported as consistent with both.

Here too the existing evidence does not decide between the accounts. Eight- and ten-year-olds gained the same speed-up from an appropriate connective in self-paced reading, with the age interaction explicitly null across 91 children (Cain & Nash, 2011). No difference in online sensitivity to a coherence break was found between children of 8 to 9 and children of 10 to 11 (Helder et al., 2016). Among 8- to 10-year-olds reading two-clause sentences, the absence of the Dutch causal connective lengthened processing only for second-language readers with weaker syntactic knowledge (Van den Bosch et al., 2018), a contrast between language backgrounds that leaves age untested, in which the connective was worth most to the reader with least resource. In fourth-graders, the effect of cohesion turned on world knowledge and far less on decoding skill, with age untestable within a single grade (McNamara et al., 2011). In the studies located for this age range, the outcome measured under a whole-text coherence manipulation is comprehension or inference, so none of them bears directly on reading rate. The three outcomes are read as for Hypothesis 2. Because the sign of a coherence effect also depends on which cohesive device carries it, connectives enter by relation type and the component's anchors are the causal and contrastive connectives. The facilitation a connective affords was smaller for result relations than for concessive ones and differed between an analytic and a synthetic language (Marchal et al., 2025), which supports both halves of that decision.

Both age-band interactions are averaged over ten languages whose orthographies differ in depth, and the band does not mean the same thing in all of them. Children in most European orthographies read familiar words accurately and fluently before the end of the first school year, while the rate of development in English is more than twice as slow, with French and Portuguese also lagging (Seymour et al., 2003). Psycholinguistic grain size theory derives that pattern from how consistently a writing system represents phonology, which sets the size of the units a learner comes to rely on (Ziegler & Goswami, 2005). Children matched on years of instruction show orthography-specific length and frequency signatures, English qualitatively unlike German and Finnish, while adults converge (Schroeder et al., 2022). Between two transparent orthographies the developmental trajectory can nonetheless be comparable, with language-specific effects confined to morphological complexity (Marzi et al., 2026), so the heterogeneity expected here is a matter of degree. Children reading different orthographies therefore sit at different points on the automatisation curve, and on the bottleneck-release reading the developmental signature would be largest where decoding automatises latest. The registered coefficient is a prevalence-weighted average across those positions, and we do not generalise it from any one orthography (Share, 2008).

Hypothesis 4: the selectivity of narration. We predict that the LexicalDifficulty coefficient will be proportionally smaller under narration than the ReferentialContinuity coefficient is. Writing π_{lex} and π_{cont} for the proportional attenuation of each coefficient under narration, computed draw by draw from the posterior under the rule given under Confirmatory models, the prediction is that $\pi_{\text{lex}} - \pi_{\text{cont}} > 0.20$. A spoken rendition supplies word identity without print-to-sound mapping, the step that a low-frequency or long word makes expensive. It does not supply the link between two ideas, and the reader must still construct that link. A voice also carries prosodic phrasing, which a silent reader must project for themselves (Breen, 2014) and which is still developing across the primary years (Godde et al., 2020). Narration could therefore lower the grammatical coefficient by a route unrelated to word identity, whereas prosodic phrasing does not directly supply the link between two sentences, which keeps the lexical-against-continuity contrast interpretable. The grammatical-by-narration interaction is estimated in the same model and will be reported for description, and it enters no decision rule. The sparing is conditional on timing. In reading while listening, adults' eyes generally ran ahead of the audio and rarely rested on the word being heard (Conklin et al., 2020), and among 119 learners of English the multimodal condition ran slightly slower than reading alone (Malone et al., 2025). Children of 8 to 9 pattern the other way (A. Valentini et al., 2024). The decoding account therefore describes the reader slower than the recording and the pacing account the reader faster than it, and the full-sample contrast reports which regime dominates the narrated sessions. The benefit of text-to-speech is largest for children whose decoding is weak but whose language is intact (Keelor et al., 2023), and in 34 children aged 8 to 9 the time saved fell on the first encounter with an orthographically novel word and had gone by the second (A. Valentini et al., 2024).

The competing account holds that narrated reading time is set by the duration of the recording, in which case each narrated slope converges on the slope of the recording's duration per word. That quantity is near zero for referential continuity and positive for lexical properties tied to word length, so the two attenuations are equal only once the recording's duration is accounted for, and without that adjustment the pacing account predicts $\pi_{\text{lex}} - \pi_{\text{cont}}$ at or below zero. Reading aloud runs at about 183 words per minute against 238 to 260 silent, and the gap between text types is predicted by word length (Brysbaert, 2019). Pacing alone also fails to flatten word-level effects, since manipulating reading speed left the influence of word properties on eye movements in place (Schwalm et al., 2026), which is one reason the registered contrast sets the lexical attenuation against the continuity attenuation. The compression diagnostic withholds the full-sample verdict where the recording sets the pace for most narrated sessions, and an adjustment for the recording's duration covers the mixture that diagnostic can miss. That adjustment rests on a predicted spoken duration per word, computed from the held texts before any outcome is joined and defined under Materials, and the refit that carries it is registered under Sensitivity analyses.

That account has substantial support of its own. The aggregate comprehension benefit of reading while listening is confined to studies whose comparison condition was externally paced (Clinton-Lisell, 2023). A registered report testing the resource-freeing mechanism in 86 adult second-language readers found comprehension *worse* under narration than in silent reading, and found that individual differences in orthographic decoding and speech segmentation did not interact with input condition in the way that account predicts (Hui & Godfroid, 2026). In 21 secondary-school students with dyslexia and 22 without, audio support increased reading time in most tasks while leaving comprehension unchanged (Knoop-van Campen et al., 2021). Where the compression has been measured incidentally

it is severe, since the standard deviation of exposure time fell from 11.56 seconds when self-paced to 0.62 seconds when narrated (A. Valentini et al., 2024). The tests located bear on comprehension outcomes and on the variance of exposure time, not on the contrast Hypothesis 4 registers.

A third possibility shares the pacing account's predictions, that the child listens and does not read, so that the interval is the recording plus the page turns. Attention to print is low in preschoolers listening to a storybook (Evans & Saint-Aubin, 2005) and rises across the primary grades with reading skill (Roy-Charland et al., 2007). In older children reading an illustrated story in a second language, audio moved the eyes from the text to the pictures without changing comprehension (Pellicer-Sánchez et al., 2020; Serrano & Pellicer-Sánchez, 2022), whereas 8- to 9-year-olds did fixate the words under bimodal presentation (A. Valentini et al., 2024). The share of narrated sessions that were read along is therefore unknown and is likely to differ by age band. The logs cannot separate a paced reading from a listening, so an inside-region verdict is reported as time set by the recording, whether or not the child read along, and never as evidence that narrated reading was reading. The narrated sessions the audio cannot have paced are the only part of the design that excludes listening by construction.

Four outcomes are registered, with the posterior criteria given under Decision rules. A difference above 0.20 supports selective removal of decoding cost. A difference practically equivalent to zero supports the account on which the recording sets the pace, with the duration-adjusted contrast reported beside it. A difference determinately below -0.05, meaning that narration removes more of the continuity cost than of the lexical cost, is consistent either with reverse selectivity, which contradicts both accounts, or with pacing acting through the recording's duration, which lengthens the audio for texts with longer words. The duration-adjusted sensitivity contrast is reported beside it as the quantity that separates the two. Anything else is inconclusive and both accounts remain live. The threshold of 0.20 is four times the region's half-width and is a registered convention.

Who reads without narration is itself a selection. Narration was on in about nine sessions in ten of the structural sample, so the unnarrated slopes, which are the denominators of both attenuation proportions, are estimated largely from the families who chose to switch the audio off. If those are disproportionately the fluent readers, their slopes are attenuated by selection and both proportions are distorted, and the data carry no measure of reading skill with which to adjust for this. We will therefore report narration prevalence by age band, reading level and language before fitting, and the rule under Confirmatory models promotes the within-account contrast to the primary estimate where the data allow it. Because the contrast is a ratio, it is evaluated only when the unnarrated slopes of the lexical and continuity components are both practically non-negligible. If either is not, there is no attenuation to compare and Hypothesis 4 is reported as not evaluable. This gating rule does not depend on the sign of any coefficient.

Hypothesis 5: the child's own judgement of difficulty. We predict that, in an ordinal model of the post-reading rating, LexicalDifficulty and GrammaticalLoad will be associated with higher odds of a harder rating and ReferentialContinuity with lower odds, each by an odds ratio outside the region. We further predict that the three standardised coefficients will rank by magnitude as they do in the reading-time model. Two measures of the same three constructs, one behavioural and one judgement-based, obtained on different scales from overlapping sets of children, are not bound to rank those constructs the same way, so agreement on order is a substantive prediction of the effort account, which measuring effort twice would deliver for free.

The link between the two measures rests on the effort heuristic of the cue-utilisation account, on which a judgement of difficulty is inferred from cues, experienced effort among them, and is not a direct reading of processing (Koriat, 1997; Undorf & Erdfelder, 2011). Two meta-analyses put the association between experienced effort and monitoring judgements at r = -.36 and, in a meta-analytic structural model, at β = -.19 (Baars et al., 2020; David et al., 2024). Those judgements are judgements of learning and of confidence, so for a difficulty rating the sign is the mirror image. Second, third and fifth graders were less confident in an answer the longer they had taken to choose it, and reliance on that cue was stronger in the older children (Koriat & Ackerman, 2010). On the proposition the theory object records, the rating is not a second instrument trained on the text but a downstream report of the same expended effort that the reading interval times. Effort is attributed to the material when the material imposes it, and effort a learner chooses to invest carries no such attribution, a distinction established in adult learners (Koriat et al., 2006; Koriat et al., 2014). In children, the association duly weakens where regulation is goal driven (Baars et al., 2020).

Reading in this app is self-paced and voluntary, which is one reason the rating is registered here as a measure under test. It is also why the theory object marks this prediction as diagnostic against the rival on which time on a text indexes how engaging the story was. Convergence in the predicted directions therefore weighs against that rival, and its absence is read under the rule registered in Table 1, as a failure of the rating and not as support for the rival.

The cue-utilisation account also supplies a reason the two measures might diverge, because a judgement made after finishing draws on the cues still accessible at that moment and the three sources of demand are not equally accessible. Word-level difficulty recurs at every line, whereas the cost of an unstated link between ideas is spread over the text, and a cohesion manipulation that moved paragraph-level judgements left whole-text judgements untouched unless readers could still consult the text (Lefèvre & Lories, 2004). The evidence is not one-sided. In adults predicting their own recall, judgements about a whole text did rise with coherence (Rawson & Dunlosky, 2002). We will report the posterior probability that the two measures rank the components alike as the study's principal convergent-validity check. At least 0.90 counts as agreement on order, and anything lower as order not established.

Two facts about the rating constrain the design. The first is that the nearest evidence is adjacent to what this rating asks. Among children of 8 to 10 solving arithmetic problems, the mental effort they reported investing was related to their judgements of their own learning, which tracked the complexity of the problem (Baars et al., 2018). Children in both of this study's bands judge how well they have understood a text (Steiner et al., 2020), fifth graders do so both for the whole text and for individual answers (Markovich et al., 2026), and primary school children are more overconfident in those judgements than secondary school children are (Mirandola et al., 2018). That overconfidence depends on the text: among 407 primary and secondary students, the primary children overestimated their performance on expository passages and inferential questions but calibrated as well as the older students on narrative texts (Zagato et al., 2026), and every text in this study is narrative. Those are judgements about the reader's own comprehension, whereas the rating here asks about the text. In adults, subjective comprehensibility ratings of narrative text can be predicted from computed text features (Tilmatine et al., 2024). The child metacognition literature located rates adjacent things, and none of it validates a global post-reading judgement of a text's difficulty against the text's measured properties in children of this age. We likewise found no psychometric evaluation of the three-point

easy, just right and difficult format, which is a classroom heuristic.

The second is that children's monitoring judgements are internally consistent when aggregated over items, their reliability is bounded by the reliability of the first-order task, and their accuracy varies from one task to another (Eberhart et al., 2025; Roebers et al., 2021). The design therefore aggregates ratings at text level and applies the five-rater floor. Children of 6 to 13 have a limited grasp of multi-point response formats, word-anchored options agreeing better than numeric ones with a yes-or-no standard (Mellor & Moore, 2014), and the number of options bears on how reliably children and young adolescents answer, four options proving optimal and seven or more lowering reliability (Borgers et al., 2004). Which way any of this would move a three-point difficulty rating inside an app is not established, so we register the direction as unknown and report the observed distribution. The identification of a reported difficulty with expended effort is itself contestable, since in 80 children aged 8 to 12 the difficulty and effort ratings given after a battery of cognitive tasks were uncorrelated with each other and only weakly related to performance (Torres & Toplak, 2026).

Judging a task by the effort it cost is itself something children come to do. In self-paced learning, judgements of learning fell as study time rose for third to sixth graders, aged 9 to 12, and showed no such relation for first and second graders, aged 7 and 8 (Koriat et al., 2009), a division that falls on this study's two bands. None of this settles the matter for a direct judgement of a text, since even 5- and 6-year-olds use an item's difficulty adaptively when evaluating their own performance (van Loon et al., 2017), so a rating that tracks the text at 7 to 8 remains possible. The rating model carries the same component-by-age-band terms as the timing model, and we will report the band-specific odds ratios beside the pooled ones. They are descriptive, and the registered tests remain the pooled odds ratios and the ordering. The recorded bands in the structural sample skewed towards the younger of the two (Supplementary Section S8), so a prevalence-weighted pooled odds ratio will be weighted towards it. Where a pooled odds ratio lies inside the region and the older band's odds ratio lies outside it, we will report that pattern as a limit on the rating specific to the younger band, and the pooled verdict registered in Table 1 stands as written.

The effective sample size is the number of distinct children rating each text, and the raw count of ratings overstates it, so the model carries a text varying intercept, and we will report the distribution of raters per text before fitting, applying the five-rater floor registered under Confirmatory models. We will report the intraclass correlation for the rating at text level, a reliability estimate for this format. We will also report the distribution of the three rating categories overall and by age band and language before fitting, and beside any inside-region verdict, because a distribution concentrated in one category limits what a cumulative logit can resolve.

For interpretation, a failure here is ambiguous on its own, and Hypothesis 1 resolves it. If the text components predict reading time but their odds ratios for the rating lie inside the region, the feature pipeline is sound and the rating is not a usable measure of experienced difficulty, and the ordering is then not interpreted. If the odds ratios lie outside the region in the predicted directions but the ordering probability falls below 0.90, the convergence is partial, and the rating is usable as a difficulty signal though not as a rank measure. The mismatch is then reported as evidence about which cues children use, leaving the components themselves untouched. If the components predict neither outcome, the components themselves are suspect. If they predict both and in the same order, two readouts of the same expended effort, reached by different routes, converge on the

same ordering of text demands. This hypothesis is also the one the design analysis marks as most likely to be underpowered. An inconclusive verdict at the census is itself informative, because it bounds the reliability a single-item difficulty check can have at the availability such apps achieve. In that case the ordering is reported descriptively and convergent validity is recorded as not established, so that the fifth outcome-neutral check has a defined outcome whether or not Hypothesis 5 reaches a verdict.

Hypothesis 6: accumulated reading and pace. We predict that, among children with several recorded sessions, a one-standard-deviation within-child increase in a child's own accumulated in-app reading, on the log scale on which it is entered, will be associated with a reduction in that child's log reading time per word. The reduction is predicted over and above session order, the interval since the previous session and the time elapsed since the child's first session. The unit is the standard deviation because that is the unit in which the region of practical equivalence is defined under Decision rules, and the implied change per doubling of accumulated words will be reported beside the coefficient.

On the orthographic-learning account, each successful encounter with a written word form is an opportunity to acquire or sharpen the word-specific orthographic representation supporting its later recognition (Share, 1995). The quality of those representations makes skilled word reading cheap (Perfetti, 2007), and experience of print is accordingly treated as the engine of growth in word reading (Castles et al., 2018; Deacon et al., 2024; Nation, 2017). Because the mechanism operates on items, the dose is counted in words of completed distinct stories, and repeated readings are held back for the sensitivity analysis registered below. Because each further form adds less as the lexicon fills, the dose enters on a log scale.

Learning of this kind is in the first place item-based. Children in grades 1 and 2 who met novel spellings in short stories learned them, but transferred that learning clearly only to orthographically related forms (Tucker & Deacon, 2025), and across 62 experiments the outcome is modulated by language-specific word properties (Li & Wang, 2023). Reading one story can therefore make the next cheaper by two routes, the words the two texts share and the sublexical regularities a reader generalises from them (Castles et al., 2018), so how far the association reaches is bounded by how much of a new text earlier reading can already have discounted. Both bounds are looser for stories written to one reading level than for texts sampled at large, and neither is measured here. Grades 1 and 2 sit below this study's bands. Outside English, orthographic learning, lexical knowledge, reading fluency and spelling were associated in German-speaking primary school children (Schmalz et al., 2025). The account is not neutral between the ten languages. Decoding automatises at very different ages across them, as the paragraph on orthographic depth above records (Seymour et al., 2003). On a model in which decipherability precedes unitisation and automatisation (Share, 2025), the per-unit speed-up that accumulated reading affords should therefore be shallower where decoding was automatised earlier. The pooled deviation coefficient remains the registered test, and the per-language slopes from the secondary refit registered under Sensitivity analyses are reported descriptively beside it, annotated with orthographic depth for the eight sampled languages Seymour and colleagues covered, with Polish and Korean reported without an annotation.

The relation between practice and skill is reciprocal over years (Torppa et al., 2020; van Bergen et al., 2021), so the coefficient is an association within the observation window, and never an estimate of a practice effect. Random-intercept cross-lagged models on more than 5,100 German secondary students found no within-person evidence that reading more yields better reading competence, while

the reverse path, from competence to exposure, remained (Bittmann, 2025). Only a design that holds the two apart can show such an asymmetry, and this one holds them apart by construction. On the other arm, exposure to a text experimentally speeded skilled adults' later recognition of the low-frequency words in it (Chetail, 2024), which is the closest experimental demonstration located, in adults and confined to rare words. That reciprocity has been established at yearly lags, so it does not by itself license treating the reverse path as negligible over a window of months. A child whose session went slowly may not return, so later sessions would be contributed disproportionately by children whose earlier sessions went well, which would push the coefficient in the predicted direction. A child reading faster may also choose longer stories, or a family may raise the reading level, and either change alters the composition of what accumulates even though the current story's length and level are adjusted for. Before the coefficient is interpreted we will report the distribution of each account's final-session pace relative to that account's own mean, and the number of accounts whose reading level changed within the window, which the audit establishes in any case. Two refits will be reported beside the registered term and will gate nothing: one excluding every account's final eligible session, and one adding the previous session's standardised pace as a covariate, in the spirit of the random-intercept cross-lagged model (Hamaker et al., 2015). The second is read with the caveat that conditioning on a lagged outcome alongside an account intercept is itself biased in short series, so it bounds the concern without settling it.

The account-mean term that enters alongside the deviation is reported but is not a test of either account. Both predict it negative, since a child who has read more in the app may read faster for having done so and a child who reads faster may go on reading. How much and how well children read have long been associated at the between-person level (Mol & Bus, 2011). Only the within-child deviation separates the accounts, and the rival has content of its own. In three twin samples, the modelled direction of causation ran from reading ability, literacy skill or comprehension to print exposure or literacy enjoyment, with the reverse influence zero in one sample (Erbeli et al., 2020; van Bergen et al., 2018; van Bergen et al., 2023).

Each covariate is there for a named alternative. Session order absorbs familiarity with the interface and with the generator's story schemas, since a reader who knows the shape of a story reads it faster for reasons the orthographic-learning account does not claim. Days since the previous session absorbs spacing. Days since the first session absorbs maturation and schooling. What identifies the accumulated-words deviation once the three are held is within-child variation in the lengths of the stories already read, which is thin where the reading level fixes story length, and that thinness is one reason the tier-2 requirement is re-derived at the observed distribution before fitting. Oral reading fluency grows appreciably and unevenly over a school year (Christ et al., 2010), so an elapsed-time covariate absorbs maturation only if growth is close to linear in days within a child and the window's position in the school year is immaterial. Growth rates also differ between children, and their predictors (Khanolainen et al., 2024) are among the things that also predict how much a child reads. Before Hypothesis 6 is interpreted we will report the within-child correlation matrix of the four covariates and the deviation term, with their variance inflation factors, and the calendar month of each session as a descriptive stratifier, so that a window spanning a school holiday is visible.

A positive within-child coefficient outside the region contradicts the orthographic-learning account and is read as within-window slowing that the covariates do not capture. A posterior inside the

region is evidence against a within-window orthographic-learning signal of at least 0.05 standard deviations. This hypothesis depends on a subsample whose size is not known before the extract is released, and it is declared underpowered, not untestable, if that subsample falls below the threshold re-derived for it under Design and precision analysis.

Every hypothesis is labelled under the single posterior rule given under Decision rules and diagnostics, and an inconclusive result is reported as such. No hypothesis is filtered by a preliminary significance test. Three labels apply before a verdict. A hypothesis is declared untestable, before its model is fitted and without reference to any coefficient, if its component fails the construction rule set out under the dimension reduction or if its outcome or contrast is absent from the frozen extract. A hypothesis whose data fall below its precision threshold in Table 1, as re-derived under Design and precision analysis where that section says so, is instead declared underpowered. Its model is still fitted and reported, and no conclusion is drawn from an interval the design analysis predicted would be indeterminate. A hypothesis is reported as not evaluable when a registered post-fitting contingency removes its test, which happens in two cases only: the Hypothesis 4 gate, and failure of the positive control on the timing column.

The limits of what follows should be equally clear. This study will not identify a causal effect of any text property, age band or narration setting. It will not estimate how fast children in general read, nor generalise beyond families who chose to install and use this application. It will say nothing about comprehension, since none of the contingent hypotheses is activated at submission. The constraints on generality are those stated here: children aged 7 and over who read narrative texts of a few hundred words on a touchscreen, in one of ten languages, inside one commercial application whose texts one generator produced (Simons et al., 2017). The medium is part of that boundary, and it is not neutral. Reading on screen yields poorer comprehension than reading the same text on paper, with the shortfall larger under time pressure and for informational text (Delgado et al., 2018), and in children the paper advantage holds except where a digital text adds features aligned with the content (Furenes et al., 2021). A network meta-analysis of 56 samples locates most of that shortfall in scrolling: where scrolling was required, paper beat every device by 0.35 to 0.48 standard deviations, and where it was not, the differences ran from 0.03 to 0.12 and were unreliable, with a tablet read without scrolling ranked closest to paper (Clinton-Lisell & Litzinger, 2026). This app serves a story as pages on a tablet, so the largest known moderator of the screen shortfall does not apply here. The laboratory results the three components are built on were obtained largely on paper, so the comparison drawn here is across media as well as across settings, and a coefficient smaller than its laboratory counterpart has that second explanation available alongside the first. Where the data controller’s answers falsify an auxiliary assumption, the affected predictions are reassigned by the rules in Table 2, never reinterpreted after the fact.

Methods

Ethics information

This research complies with all relevant ethical regulations. Data are generated in the ordinary operation of a commercial application and are collected by its publisher under the terms of use and privacy policy a family accepts on installation, with consent for research use of anonymised records registered in the app by a parent or guardian. That registration is the informed consent under which the records reach us, and no further consent is sought. We play no part in collection, have no contact of any kind with any child or family, and return nothing to the platform, so nothing done here can alter what any child experiences. No compensation is provided or applicable. The extract supplied to us carries no direct identifier: the account key is a pseudonymous string generated by the platform, and no name, contact detail, precise location or device identifier is included. Ethical approval for secondary analysis of these anonymised records [has been granted by / is being sought from] the Departmental Research Ethics Committee of the Department of Education, University of Oxford (reference [to be inserted]), and will be in place before acceptance in principle. The study involves no experimental manipulation and no intervention, so it raises no question of withdrawal from a procedure. A family's right to have its records erased is exercised through the platform, and any erasure occurring between the building of the extract and its transfer will be recorded by the data controller and reported at Stage 2.

Two conventions hold throughout this section. First, no decision about whether a hypothesis is tested, on which sample, or with which model, depends on a fitted coefficient. Every such decision follows from provider-computed counts, the data controller's written answers, or thresholds registered here, with one exception, the Hypothesis 4 gate. Second, wherever the text says a quantity will be reported, we will report it in the Stage 2 manuscript whether or not it changes a conclusion. Two of the authors of this protocol have published a descriptive study of this platform for an earlier thirty-day window, working only from summaries the data controller prepared (Ratner & Murphy, 2026). That study relates no text property to any behaviour and analyses no reading duration, and nothing it reports is an input to any decision registered in this protocol.

Pilot data

The pilot material is a 35-row structural sample of the session table, the data dictionary, and a summary of that sample computed with the count script described under Data provenance. The first two were received from the data controller on 27 July 2026 to confirm the shape of the table, and the third we computed from the sample on 6 August 2026. The sample carries real reading times for 35 sessions on five distinct texts, one session per account, in ten reading languages. It cannot support any hypothesis in this protocol: none is identified by 35 observations, and no within-child term is estimable from one session per account. It fixed three inputs of the design analysis, which are the narration prevalence (on in 32 of 35 sessions), the rating availability (present in 10 of 35) and the age-band missingness (blank in 10 of 35), and Supplementary Section S8 reports it in full. The sample is deposited for peer review under embargo until Stage 2 acceptance, at the location the Data availability statement names.

Design

Data provenance, sequestration and outcome blindness

The data this protocol analyses already exist. Nature Human Behaviour admits secondary analyses of existing data on condition that authors can evidence no prior access to the data in question. We put the arrangement set out below to the journal's editorial office, which confirmed it as appropriate on 3 August 2026, before we wrote this protocol.

The data controller builds the export now, on its own release cycle, and holds it. Nothing of the session-level file reaches us until the journal confirms in-principle acceptance in writing. In the interval, three things may pass to us, each of which the editorial office agreed may be received in advance. The first is a manifest: the column list, the row count, the cut-off timestamp, the build date and a SHA-256 checksum for every file, so that the file eventually transferred can be shown to be the file that was built. The second is the story texts, keyed by story identifier, reading language, reading level and app version, with page breaks preserved, because every linguistic measure in this protocol is computed on them and none of them contains a behavioural outcome. The third is a set of aggregate counts computed by the data controller from the export by running a script we supplied, which reads nineteen columns besides the outcome, among them identifiers, the session timestamp and page counts, and the reading-time column, writes one small table of counts and a page drawn from it, and never reports a reading time as a value. Both the version sent on 7 August 2026 and a revised version, which adds checks against malformed input without changing what is counted, are deposited with this protocol; the counts will be computed with the revised version, sent on [date]. At submission we hold the story texts, received on [date]. The manifest and the provider-computed counts [had been received on DATE / had not yet been received], and the count of distinct texts with eligible readings in the extract, which only those counts settle, is reported in Table 2 as unresolved.

A count computed from an outcome column is where blindness would fail. The script treats reading time only as a column whose usability has to be established. It reports how many values are absent, non-numeric, negative or zero, and how many fall into each of seven bands whose boundaries are successive powers of ten, together with the number of distinct values the column holds and the largest number of sessions sharing any single value. It never crosses reading time with language, age, text, account or date. Bands ten times wider than one another cannot distinguish a fast text from a slow one. What they can distinguish is a column recorded in seconds from one recorded in milliseconds, which the analysis has to know before it can be specified at all. The count of sessions sharing one value is how a session-timeout cap makes itself visible, and the sampling plan below turns that count into a registered decision about censoring.

The structural sample and the summary computed on it, described under Pilot data, are part of the record. At the editorial office's specific direction, we have not run the dimension reduction specified below and will not run it before submission, although we hold the texts on which it would operate. The component solution is therefore fully specified here and entirely unfitted.

The data controller will supply a signed certification naming an owner and a deputy for the held file, and stating its build date, its window, its row count and its checksums. The same document certifies that no part of the file has been transferred to the research team or to anyone acting for it, that transfer will occur only on written confirmation of in-principle acceptance, and that no analysis relating story text to reading time has been conducted by the controller or reported to the team. It is

reproduced in the supplementary material.

Materials

The reading materials are the story texts themselves, each written by a generative model around the combination of five attributes a family chooses. A text is then fixed, and it is not rewritten when the same combination is chosen again. Texts also exist in a fixed classic set that families may read instead of a generated story. Alongside the texts, the extract records which of the two a reading drew on, the reading level the platform was serving, the language, the app version and the page count. Every session will be joined to the features of the exact version it was served, matched on story identifier, reading language, reading level and app version.

For identification, a text is a stimulus shared by many readers, and every text coefficient in this protocol is identified by comparing texts. How many distinct texts carry eligible readings is therefore the single quantity that governs how much this design can establish, and it is not yet known. The data controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes maps to one text shared by everyone who picks it or to a separate text per session. The published description of the platform is the prior on both questions: a catalogue of over 40,000 English-language stories generated from the element combinations, many of them translated into the child's home language (Ratner et al., 2025). That figure is the catalogue as its authors describe it, and the same paper reports over 2.4 million story sessions in the first six months after launch. If that description holds for the window, texts are shared across sessions and differ across languages by translation, which would trigger the varying intercept for the underlying story crossed with Language that Table 2 registers under question (d). The controller's written answer supersedes it. Table 2 records what follows from each answer.

For every text, we will compute features in four families. *Lexical* features are mean log frequency of content words, the proportion of words with a frequency below one per million tokens on the uniform frequency resource, mean rated age of acquisition, mean orthographic length, mean per-word surprisal under a language-specific neural language model, lexical density, lexical diversity and concreteness. Diversity is the moving-average type-token ratio over a 50-token window, with the Measure of Textual Lexical Diversity computed beside it and the per-language correlation between the two reported. A word's surprisal is the sum of the surprisals of the subword tokens composing it, and the mean is taken over the same whitespace-delimited words that form the outcome's denominator. *Grammatical* features are mean dependency distance from a universal-dependencies parse, clauses per sentence, the ratio of subordinate to main clauses, mean sentence length and mean depth of noun-phrase modification. *Discourse* features are mean semantic similarity between adjacent sentences from a multilingual sentence encoder, referential overlap, connectives per hundred words by relation type and narrativity. Referential overlap is the proportion of adjacent sentence pairs sharing at least one content lemma in a nominal position, which implements the argument-overlap rule of the construction-integration model without a coreference resolver. Narrativity is a stipulated composite of past-tense marking, personal pronouns and direct speech, defined in Supplementary Table 3. *Affective* features are the mean and the variability of valence and of arousal. One further text quantity serves Hypothesis 4 alone, a predicted spoken duration per word on a log scale, computed from syllables per word together with sentence- and clause-boundary pauses per word from punctuation. Its resources and matched-token proportion are given in Supplementary Table 3. Where the data controller supplies the per-text audio duration, that replaces the prediction, under the rule given

below.

What the narration consists of is not established. The extract records only a session-level flag. The platform has not disclosed whether the voice is recorded or synthesised, how many voices serve a language, whether words or lines are highlighted in time with the voice, whether a page advances when playback ends, whether playback can be paused or repeated, or the playback rate. Two of these bear on how a narrated reading is to be read. Word-level highlighting couples the eye to the voice, though the evidence comes from children younger than any here (Liao et al., 2020). A page that advances when playback ends would make a narrated session's duration largely the duration of the audio. These items are added to question (a) in Table 2, with the consequence of each answer stated there, and any item we are able to establish from the public version of the app is reported as observed, with the date. Observing the app involves no outcome data, and none of these facts enters a model.

Supplementary Table 3 gives the primary and the sensitivity resource for each feature family, with its coverage across the ten languages the structural sample showed. The ten languages are Dutch, English, French, German, Greek, Italian, Korean, Polish, Portuguese and Spanish. Four rules govern that table and are fixed before any feature is computed. First, each feature will be computed from a single resource applied identically in every language, even where a better native resource exists in one of them, and native norms will be run as a pre-specified sensitivity analysis with the per-language correlation between the two reported. Second, no value is ever carried across languages by translating a word list, and a predicted resource is admitted only where the prediction was made within each language by one documented procedure and its correlation with native ratings is reported. Where no acceptable measure exists for a language, that language does not contribute that feature.

Third, every measure reports the proportion of tokens it matched, and a language in which a measure matches fewer than 80 per cent of content-word tokens does not contribute it. Content-word tokens are those whose universal part-of-speech tag is NOUN, VERB, ADJ or ADV. This token-coverage rule is distinct from the text-coverage rule under the dimension reduction, and the two can give opposite verdicts on the same feature. Where a feature is available in some languages and not others, that family's pooled solution is computed on the features common to every contributing language, and that feature enters the per-language descriptive solutions and the language-specific sensitivity analyses alone.

Proper nouns are excluded from that denominator and from every norm-based mean, because this platform puts invented ones into every story by construction. The hero, the companion and the setting a family chose are named in the text, and no subtitle frequency list, age-of-acquisition norm or concreteness norm contains an invented name. Because those names are chosen by the reader and recur through the story, we will also report three length-normalised name quantities computed from the same parse: the proportion of tokens tagged as proper nouns, the number of distinct proper-noun types per hundred words, and the mean repetitions per type. Each norm-based mean is reported with and without the exclusion, with the per-language correlation between the two versions. Surprisal is computed with the names present in context and excluded only from the mean.

Fourth, the full inventory, the software versions, the model checkpoints, the tokenisation rules and the matched-token proportions will be frozen and published before any feature is joined to an

outcome. Sentence segmentation is registered with them. Pages are concatenated with a single space and a page break is never a sentence boundary. Per language, we will report the mean and standard deviation of sentence length, the proportion of sentences ending without terminal punctuation, and boundary agreement between the two parsers' splitters on a stratified sample, with a native-speaker adjudication of twenty texts per language reported alongside the hand-checked parse. Korean is the language these rules constrain most, and the consequences for it are set out beside that table.

Measures

The processing outcome, serving Hypotheses 1 to 4 and 6, is the natural logarithm of elapsed reading time divided by the story's word count, standardised within reading language. A word, in that denominator and in every per-word quantity in this protocol, is a whitespace-delimited token of the text as served. In nine of the ten languages, that is a word in the ordinary sense. In Korean it is an eojeol, a content word together with its postpositional particles or verbal endings, which is also the unit the Korean dependency treebanks and both parsers use. A Korean text therefore holds fewer tokens than a translation of it would, so a Korean rate in words per minute is not comparable in content with the other nine, and the plausibility bounds below and the read-aloud fallback under Confirmatory models are correspondingly less binding for Korean at the fast end. Korean is therefore read through its own language-varying slopes throughout, as Supplementary Section S3 records for orthographic length.

Three further quantities are reported with the outcome. Where the extract carries the platform's own word count, we will report its agreement with ours before any outcome model is fitted. We will also report the intraclass correlation of the outcome at the Text and at the Account level from the tier-1 variance components, as the reliability with which a text and an account are characterised by their readings. Mean orthographic length anchors LexicalDifficulty and also helps determine how many words a text holds, since for the same content a text of longer words holds fewer of them, so time per word rises with word length even where time per character is unchanged. A registered sensitivity analysis, listed below, closes that path, leaving the primary scale as it stands. We describe it throughout as reading pace, an index of the processing demand a text imposes under the auxiliary assumption stated in the Introduction, and never as a measure of comprehension or fluency, because nothing in these data licenses that reading. The outcome for Hypothesis 5 is the child's optional post-reading rating of the story as easy, just right or difficult, treated as ordered and modelled on a cumulative-logit scale.

The confirmatory predictors are the three component scores defined below. Text length in words, narrativity, mean valence and mean arousal are pre-specified secondary text predictors and enter every model as covariates. Valence and arousal effects on word-recognition latency are larger among low-frequency words (Kuperman et al., 2014), so the affective means overlap LexicalDifficulty and enter as adjustments and never as tests. Omitted, they would leave part of that component's association unadjusted. No interaction between an affective mean and a component is estimated, and the overlap is made visible by the variance inflation factors registered under Decision rules. Every affective value used here descends from adult ratings, which is a limitation Supplementary Section S3 records, so these covariates are read as properties of the text on an adult referent and never as what a child felt. The variability of valence and of arousal, and the additive and temporal connective densities, will be computed for description only: they enter no confirmatory model and are excluded from the component solution. Whether the text was generated or drawn from the

classic set enters as a covariate, and the substantive comparison between the two is a contingent hypothesis, for reasons given under Table 2. Because predictors are standardised within language, a pooled coefficient averages slopes expressed in each language's own standard-deviation units. We will report the raw-scale standard deviation of each component per language so that readers can translate, and the language-varying slopes summarise the heterogeneity this induces.

Three variables describe the reader and the reading context. The age band is the platform's own record of the child's age group, reduced to the two bands this study covers. An account whose recorded band changes within the window contributes each session under the band recorded for it, and the number of such accounts will be reported. The reading level is the level the family has set for the session, which determines which version of a text is served. Narration is the session-level flag recording that a spoken rendition accompanied the reading.

Accumulated in-app reading, which serves Hypothesis 6, is the cumulative count of words in completed distinct stories the account read before the current session, entered as $\log(1 + \text{accumulated words})$ so that the first eligible session, at which the count is zero, is defined. The transformed variable is standardised within language, as every continuous predictor is, and only then decomposed into the account's mean and the current session's deviation from it, so the registered coefficient is a change in the standardised outcome per one within-language standard deviation of that deviation. Session order, the number of days since the account's previous eligible session and the number of days since the account's first eligible session enter alongside it. The deviation term is the registered test.

The data are crossed, and not nested: readings sit within accounts, a text is met by many accounts and exists in several languages and at several reading levels, and an account may read in more than one language. We will therefore treat Account, Text, Language, Country and App version as crossed grouping factors, and report the counts supporting each before fitting any model. A grouping factor other than Text and Account enters when it has at least five levels with at least 20 eligible readings each. Below that it is dropped, and the drop will be reported. Reading level, with only three levels, enters as a fixed factor. If at least 90 per cent of texts are read in only one eligible session, a Text varying intercept is not estimable and is replaced by varying intercepts for the repeated attribute combination that generated the text. These thresholds are registered conventions, and any consequence of them will be reported.

Randomisation and blinding

Nothing in this study is randomised, and no allocation, assignment or condition is under anyone's control.

Data collection and analysis will not be performed blind to the conditions of the study. Data collection is performed by the platform without researcher involvement. Before in-principle acceptance, blinding is structural: we cannot see the outcome data, which the data controller holds under the sequestration arrangement described above. After transfer, analysis will not be blind to the values of the predictors or grouping variables, since the registered models require them. We will write, run and debug the outcome-model code against data simulated from the design specification deposited with this protocol, so that every script is complete and exercised before the real file exists on any machine of ours.

Sampling plan

The design is observational and the sample is a census: we will analyse every eligible record in one immutable extract, whose closing date is fixed before any outcome model is fitted, with no sequential testing and no optional stopping.

Two nested samples serve different hypotheses, which are identified at different levels. Tier 1 carries the text hypotheses. It comprises every completed reading in the extract for which the account is recorded in age band 7-8 or 9 and over, the reading language is one the text-feature pipeline covers, the story text has been supplied, and the reading time falls inside the plausibility bounds below. Where an account read the same text more than once, only the first reading enters tier 1, and repeated readings are retained for the practice analysis described under sensitivity. The unit of analysis is one reading of one text by one account.

If at least ten per cent of tier-1 readings come from accounts contributing two or more readings, the model includes a varying intercept for Account. If fewer do, that term is not separable from residual variance: it is dropped, child variance is carried in the residual, and the reduction in the effective independence of the sample is stated wherever a tier-1 interval is reported. The ten per cent figure is a registered convention. We will read the proportion from the provider-computed counts before fitting anything.

Tier 2 carries the within-child hypothesis. It comprises those tier-1 accounts contributing at least three eligible readings of distinct texts. Three is the smallest number that leaves a degree of freedom once a within-child slope and an intercept are estimated. How many accounts reach the tier-2 threshold is not known in advance.

Reading time enters as elapsed seconds for the whole story divided by the story's word count, and we bound it by what a child could physically be doing, and not by anything observed in these data. We will exclude a reading faster than 400 words per minute, or 0.15 seconds per word, as faster than reading. We will exclude a reading slower than 10 words per minute, or six seconds per word, as a story left open. The upper bound is the top of the adult range and follows published precedent. Silent reading in educated adults typically runs between 200 and 400 words per minute (Rayner et al., 2016), so a child of 7 to 11 sustaining more than 400 is not plausibly reading every word of the page. A meta-analysis of Spanish reading rates applied that same cut for that reason, eliminating rates above 400 words per minute as a sign that the reader had skipped part of the text, and removing 313 cases from tests that assessed 19,237 participants, every one of them a silent-reading measure (Ripoll Salceda et al., 2020). The lower bound sits below the slowest pooled mean in that meta-analysis, 30 words per minute for Spanish first-graders reading silently, though not below the floor of the normal range proposed for that grade, 8 words per minute. We will report the number and proportion removed by every rule, by language, age band, reading level and narration state.

If the provider-computed counts show more than one per cent of sessions sharing a single value, we will locate that value in the transferred extract before fitting any outcome model. If it lies in the upper decile of the reading-time distribution, we will treat it as a right-censoring point and fit the timing models as censored lognormal models, with those observations retained as censored.

Ahead of all of the above, we will exclude sessions the platform flags as incomplete or crashed, duplicated event records, accounts flagged as duplicates under the platform's own data-quality

rules, and stories without a positive word count. No excluded datum is replaced. A participant-flow diagram will report every exclusion and the retained accounts, readings and texts, overall and by language.

Measurement, availability and missingness audit

Digital traces have their own error taxonomy, distinct from that of surveys (Sen et al., 2021), and four concerns here fall under the types it names. Three are trace measurement errors, where what the platform records stands imperfectly for the construct: the platform's own age-band and reading-level categories, a timing interval whose endpoints the documentation does not fully specify (question c), and drift in the instrument as the app changes version across the window. That drift is also a change in platform affordances, which the framework names as a type in its own right. The fourth concern is a trace selection error, since a session that times out is right-censored and never reaches the extract.

Before fitting any outcome model, we will run and archive an audit of the frozen extract and its dictionary. It establishes field availability and units, the granularity of the age field, how often a text and a reading level recur, the narration flag's behaviour within accounts and within sessions, duplicate and crash flags, changes across app versions, per-language feature coverage, and every nesting and crossing count the models require. It also reports missingness for every planned variable, overall and by language, age band, reading level, narration state and session order.

A feature that is structurally unavailable in a language is not imputed: the language simply does not contribute it. A session missing an outcome is excluded from the model for that outcome alone, and a missing narration flag does not remove a session from a model that has no narration term. No variable is selected, transformed or discarded because of its association with reading time or with the rating.

The rating in Hypothesis 5 is optional and was absent for 25 of the 35 sessions in the structural sample, so high missingness is expected. Where the rating is missing for more than five per cent of otherwise eligible sessions, we will model availability from observed pre-response variables and report an inverse-probability-weighted analysis as a sensitivity check, with weight distributions and common support shown.

Design and precision analysis

Nature Human Behaviour's guidelines set 0.95 as the a priori power floor for frequentist plans and, for Bayesian plans, allow a maximum feasible sample provided an inconclusive result at that size would still be an important message. This design is neither a significance test nor a Bayes-factor test but an estimation design on a census of existing records, and it maps onto the second provision. The frozen extract is the maximum feasible sample, since no further data can be collected, and 0.90 is registered as the assurance level at which a hypothesis counts as adequately powered, with every 0.95 requirement reported beside it from the same archived curves (Supplementary Section S5). A hypothesis declared underpowered is still fitted and reported, so its interval bounds the effect even where it cannot decide it.

The text coefficients are identified between distinct texts, and how many distinct texts the extract holds is unresolved.

We therefore simulated data from the design specifications deposited alongside this protocol with

pilotr, sweeping the number of distinct texts at constant readings per text and separately sweeping readings per text at a constant number of texts (Bernabeu, 2026b). The simulated tier-1 model contained the three components, the age band, the two age-band interactions and a text intercept, fitted by restricted maximum likelihood with lme4. The narration model contained the lexical and continuity components, the narration flag and their interactions with it. The tier-2 model contained the three components, a single accumulated-reading trend and child and text intercepts, without the age band or its interactions, and each simulated child read five distinct texts, one at each session position. Simulation was on the within-language standardised scale, with the text intercept contributing 9 per cent of the variance, the child intercept 20 per cent and the residual the remainder, so a simulated coefficient is directly comparable with the region of practical equivalence. We ran the sweep at twice the region's size, the smallest effect this protocol treats as worth reporting, and a null condition with every coefficient set to zero established how reliably the design can declare practical equivalence.

These simulated models are simpler than the registered ones, and the simplifications do not all push the same way. They omit the narration terms of the tier-1 model, the reading-level adjustment, the covariates, the language, country and app-version intercepts, the language-varying slopes, the distributional sub-model and the review-derived priors. The simulation also generated child variance and then handed the analysis a dataset in which each account contributes one reading, so that variance reached the fitted model as inflated residual; in that one respect the requirements below for the reading-time models are an upper bound on what the design needs. The same omission does more in the ordinal model for Hypothesis 5, where it also attenuates every coefficient towards zero, so that projection is computed from the attenuated estimate and carries no such bound. Adjusting for reading level, which moves all three feature families together, will remove some of the components' variance and raise every requirement, whereas the language-varying slopes and the priors will lower them. The registered prevalence-weighted main effects carry the precision of the pooled slope and no design effect, whereas each interaction inherits the design effect for imbalance that the adjustments below apply. We will refit one grid point of each sweep with the registered brms specification when the design analysis is re-run at the observed inputs, and report the difference.

The criteria used to score each simulated verdict are slightly stricter than the rules registered under Decision rules and diagnostics. The simulation counted a verdict as determinate when the 95 per cent interval lay wholly outside the region, and equivalence when it lay wholly inside. An interval wholly outside the region leaves at most 2.5 per cent of a symmetric posterior inside it, where the registered rule allows up to 5 per cent, and an interval wholly inside the region guarantees at least 95 per cent of the posterior inside it without being required for it. The requirements below are therefore conservative with respect to the registered decision rules.

The region of 0.05 standard deviations and the 0.10-standard-deviation effect are registered conventions, the second being twice the first because an effect at the boundary can never be declared outside it. The prior-derivation pipeline (Supplementary Section S6) will record, before any outcome datum is accessed, whether each component's pooled estimate on the model's scale exceeds 0.10 standard deviations. Where a pooled estimate lies between 0.05 and 0.10, the texts sweep will be re-simulated with the deposited scripts at that effect, replacing the single 0.10 input, and the resulting threshold replaces the one in Table 1 for that component before fitting. Where a pooled estimate lies at or below the region's half-width, no number of texts could yield a determinate verdict at that size. The

0.10 convention then stands, and the discrepancy will be reported beside the component's verdict.

For the three text coefficients, an effect of 0.10 standard deviations reached a determinate verdict nine times in ten at 738 distinct texts for lexical difficulty (95 per cent interval 640 to 843), 660 for grammatical load (593 to 731) and 634 for referential continuity (568 to 704), and 95 times in 100 at 926, 812 and 784. The interactions with the age band needed roughly twice as many: 1,230 texts for grammatical load (1,107 to 1,359) and 1,431 for referential continuity (1,276 to 1,596) at 0.90, and 1,510 for grammatical load at 0.95, a target the continuity interaction did not reach within the 1,600 texts simulated. The narration contrast needed 1,234 texts (1,113 to 1,361) before the design effect for imbalance is applied, and 1,506 at 0.95. That figure prices the lexical-by-narration interaction coefficient, and is a lower bound for the difference-of-proportions rule Hypothesis 4 registers, whose draw-by-draw ratio has heavier tails than the coefficient it is built from. Declaring practical equivalence when there is nothing there cost about the same: at 800 texts a main effect was correctly declared equivalent in 87 to 90 per cent of replicates and at 1,600 in every one, while the interactions reached 89 to 92 per cent only at 1,600, over 120 replicates. An inconclusive verdict is therefore predictable from the number of texts alone. The within-child term needed 609 children in tier 2 (528 to 697), each reading five distinct texts, and 731 (621 to 851) at 0.95, for the unadjusted trend. The registered term is adjusted for session order and elapsed time, with which accumulated words are correlated within a child, so this figure understates the requirement, and the tier-2 re-run described below re-derives it with those covariates in place.

Holding the number of distinct texts at 250 and raising readings per text from 2 to 60 narrowed the interval on a main effect from 0.19 to 0.08 standard deviations, yet never carried it to a determinate verdict, reaching 0.70 at best. Over the same range, the age-band interactions climbed from under 0.11 to between 0.87 and 0.90. Sessions and texts are therefore not interchangeable: a text coefficient is bounded by how many different texts the platform generated, whereas an interaction between a text property and a reader attribute gains from many different children meeting the same text.

Figure 3 shows both sweeps. The threshold each hypothesis has to clear is recorded in Table 1, and we fixed it before the extract was released. A hypothesis whose required contrast is absent will be labelled untestable, and one whose simulated interval cannot separate the region's boundary from the effect will be declared underpowered, before its model is fitted.

Six adjustments of the design analysis are registered, each a function of the provider-computed counts or of the archived prior pipeline. Two are analytic. The simulation balances the narration contrast, whereas narration was on in 32 of the 35 sessions of the structural sample. An unbalanced binary contrast at prevalence p inflates the variance of its interaction coefficient by $1/(4p(1-p))$ relative to balance, so we will multiply the simulated Hypothesis 4 requirement by that factor at the observed narration prevalence and state the adjusted requirement before fitting. At a narration prevalence of 0.91, equivalently an unnarrated share of 0.09, the factor is about 3.0. The second adjustment applies the same factor to the Hypothesis 2 and 3 requirements at the observed age-band prevalence, since the simulation balanced the bands as well. Three are re-runs of the deposited scripts. The third adjustment re-runs the rating simulation at the observed rating availability, replacing that single input. The fourth re-runs it at the observed proportions of the three rating categories. Both are read from the extract itself. Figures of the same kind have been published for an earlier window of this platform (Ratner & Murphy, 2026), but they cover ages from three years below this study's youngest band, so they are not used as inputs here. The fifth re-runs the tier-2 sweep at the

observed distribution of sessions per child, and for that re-run alone the deposited tier-2 specification will be extended with session order and days since the first session, with per-session word counts generated so that accumulated words correlate with session order as the counts show. Each extended specification will be deposited before it is run. The thresholds those re-runs yield replace the ones in Table 1 for the underpowered declarations of Hypotheses 5 and 6. The sixth adjustment is the re-simulation at a pooled effect below 0.10 described above, which runs only if the prior pipeline calls for it.

Analysis plan

We will fit all confirmatory models as Bayesian hierarchical models with the brms package and its Stan backend (Bürkner, 2017), standardising continuous predictors within language after transformation. Wherever the accompanying systematic review supports it, weakly informative priors will be replaced by priors derived from that review, as described under Priors.

Outcome-blind dimension reduction

The observed text features are correlated because the same decision about what reading level a text is written for moves lexical, grammatical and discourse properties together. We will reduce them by principal component analysis without treating them as effects of an unobserved latent cause (Crossley, 2025), separately within the lexical, grammatical and discourse families. The affective measures will be kept as separate secondary predictors, because valence and arousal are theoretically distinct and need not move together. Narrativity, though computed as a discourse feature, will likewise be excluded from the component solution. Each distinct text will contribute one row however often it was read, so a popular text does not determine the solution. A text enters a family's solution only if it has valid values on every retained feature of that family. A text also enters no family's solution unless it holds at least five sentences after the segmentation rules registered under Materials are applied, which is the boundary condition the theory object records for referential continuity. Because the tier-1 model carries all three components, a text below that floor contributes to no confirmatory model, and the thresholds in Table 1 are read against the distinct-text counts after this rule is applied. We will report, per language, the distribution of sentences and of words per text and the number of texts the rule removes.

Counts will be transformed with $\log(1 + x)$ and bounded proportions with an empirical-logit transform where these reduce skew. A feature is dropped from a language if more than 20 per cent of values are missing, if fewer than 80 per cent of texts receive a valid token-level match, or if its standard deviation after transformation is below 0.01. A language contributes to a family's solution only if it has at least 100 distinct eligible texts and at least three valid features in that family, two of them anchors. Where two features correlate at $|r| > .90$, the retained one is chosen in this order: lower missingness, then broader language coverage, then higher extraction reliability on a manually checked sample, then the measure listed first under Materials. No outcome correlation enters any of these decisions.

Features will be standardised within language. To stop the largest language determining the solution, we will compute a correlation matrix within each eligible language, transform the correlations to Fisher's z, average the matrices with equal weight per language, and transform the average back before the analysis (Revelle, 2023). Sampling adequacy requires individual Kaiser-Meyer-Olkin values above .50 and an overall value above .60. The number of components will be set by parallel

analysis (Horn, 1965), retaining a component whose eigenvalue exceeds the 95th percentile of the corresponding eigenvalues from 10,000 random datasets. That percentile criterion is Glorfeld's modification of Horn's average (Glorfeld, 1995) and is the more conservative of the two. The scree plot will be reported but does not override that rule. Where a family retains more than one component, we will use promax rotation, because linguistic properties co-occur rather than varying independently (Biber, 1988), with $\kappa = 4$, the default of the implementation used (Revelle, 2023). We will then apply the regression scoring coefficients from the frozen, language-balanced solution to each text's within-language standardised features.

Each confirmatory component must load at least .30 in absolute value on at least two of its anchors. For LexicalDifficulty, the anchors are frequency, surprisal and word length. Surprisal is grouped with the lexical features because it and word frequency are hardest to hold apart at the word level: frequency compensates for surprisal to a degree that depends on the estimating model (Oh & Schuler, 2025), and the divergence between model surprisal and human reading times is concentrated on the rarest words (Oh et al., 2024). The correlation between surprisal and each grammatical feature will be reported per language beside the pattern matrix, and the family assignment fixed here does not depend on it. For GrammaticalLoad, the anchors are dependency distance, subordination, clause density, sentence length and noun-phrase depth. For ReferentialContinuity, they are adjacent-sentence similarity, referential overlap and the density of causal and contrastive connectives. Age of acquisition contributes to the lexical family where a broad native norm set exists but is not an anchor, because the cross-linguistic set holds only 299 words per language. Only the causal and contrastive connective densities enter the discourse family, and both are anchors. The additive and temporal densities are descriptive only, as stated under Measures, because causal and contrastive connectives raised comprehension, temporal ones did nothing and additive ones lowered it, so a single mixed count has no defined expected sign (Kleijn et al., 2019). Signs are fixed so that higher LexicalDifficulty and GrammaticalLoad mean a more demanding text and higher ReferentialContinuity a more explicitly connected one.

Three rules settle what follows from that threshold. If no component in a family satisfies it, the hypotheses resting on that family are reported as untestable. Where more than one component does, the confirmatory component is the one on which the largest number of anchors load at or above .30 in absolute value, and the other components are reported descriptively. In the discourse family, a tie is broken in favour of the component on which the causal and contrastive connective densities load highest, because the cue-utilisation account that Hypothesis 3 adjudicates is stated for connectives. That is a registered convention for breaking a tie, and not a claim that connectives are the more predictive cue: in the child sample closest to this one, deep cohesion and referential cohesion both carried unique comprehension variance (Pickren et al., 2022). If the component so chosen loads below .30 on both connective densities, Hypothesis 3 is still tested on it, but the interaction tests only referential and semantic continuity, and the adjudication between the knowledge and cue-utilisation accounts is reported as not addressed.

The solution is reported in full: the complete pattern matrix, the component correlations, the variance explained, the language-specific matrices, the loading congruence for the five largest languages and the raw-scale standard deviation of mean dependency distance per language. Concreteness needs a rule of its own, having a usable rated set in nine of the ten languages while not being an anchor for LexicalDifficulty. It is dropped from Korean under the missingness rule above, and from any

further language whose matched-token proportion falls below the token-coverage rule; it enters the language-averaged correlation matrix as the average over the languages retaining it; and the lexical solution will be reported both with and without it, with Tucker's coefficient of congruence between the two. Language-specific and sparse solutions are sensitivity analyses, and never alternative routes chosen after seeing an outcome.

Confirmatory models

In tier 1, we will estimate a model of within-language standardised log reading time per word. Its fixed effects are the three components, the age band, the narration flag, each component's interactions with the age band and with narration, the reading level, log text length, narrativity, valence, arousal and the generated-or-classic indicator. The age band and narration enter as deviations from their observed prevalence, as stated in the Introduction. Its grouping structure comprises varying intercepts for Text, Country and App version, varying intercepts and uncorrelated component slopes for Language, and a varying intercept for Account where the estimability rule above is met. Hypotheses 1 to 4 will be evaluated in this model. Because the model carries every such interaction, each registered interaction is estimated in a full set and each main effect is the prevalence-weighted average slope defined in the Introduction. Three of those terms carry no registered test: LexicalDifficulty × AgeBand, GrammaticalLoad × Narration and the components' remaining narration terms outside the Hypothesis 4 contrast. They will be reported with posterior medians and intervals as descriptive estimates, with no decision rule attached, so that no estimated term is reported selectively. No signed expectation is registered for the lexical interaction with the age band. The developmental evidence is expressed in raw latencies on isolated words, and a raw-scale attenuation is not a proportional one on the log scale this outcome uses. Across grades 2 to 4, word length and regularity effects declined while frequency effects persisted (Heß et al., 2026), and in Italian children of grades 1 to 3 sensitivity to word length fell with age (Zoccolotti et al., 2005), that study having manipulated length alone. None of these estimates is read as evidence for or against the assumption that the age band stands in for a bundle of developing capacities. The slope of a component in a given state is recovered draw by draw as the main effect plus the interaction multiplied by that state's deviation score.

The Hypothesis 4 contrast is computed as follows. For each posterior draw, the proportional attenuation of a component is one minus its narrated slope divided by its unnarrated slope, so that a positive value means a smaller coefficient under narration. The gate registered in the Introduction is evaluated on the derived unnarrated slopes: the contrast is computed only when fewer than 5 per cent of the draws of each unnarrated slope lie inside the region of practical equivalence. Draws in which either unnarrated slope lies inside the region will be excluded from the ratio, and the excluded fraction will be reported. As a registered sensitivity analysis, the contrast will be repeated with a fixed denominator, each narration interaction divided by the posterior median of its unnarrated slope, and reported beside the draw-by-draw version.

Three further rules govern Hypothesis 4. First, the within-account contrast, restricted to accounts observed in both narration states, becomes the primary Hypothesis 4 estimate if those accounts cover at least as many distinct texts as the adjusted requirement, with the full-sample contrast reported beside it. Otherwise, the within-account contrast remains a sensitivity analysis, and the selection into narration state is named wherever the full-sample estimate is reported.

Second, the tier-1 model carries a distributional sub-model in which the residual standard deviation

is itself predicted by the narration flag, and the ratio of the narrated to the unnarrated residual standard deviation is the compression diagnostic. A ratio below 0.5 is the registered signature of a recording setting the pace. If it fires, the full-sample contrast is not interpreted and Hypothesis 4 rests on the subset described next. Beside the ratio we will report, for each language, the share of narrated sessions whose rate lies within 10 per cent of that language's median narrated rate, which shows a minority of sessions sitting on the recording that the ratio can miss. This share is descriptive and gates nothing.

Third, the contrast will be repeated in the narrated sessions the audio cannot have paced. For each language, the narration rate will be estimated as the median words per minute of its narrated sessions, and a narrated session counts as unpaced when its rate exceeds 1.5 times that median. Where a language has fewer than 100 narrated sessions, a fixed bound applies instead: that language's own adult read-aloud mean in the International Reading Speed Texts where the language is among the 17 covered, and otherwise the slowest of those means, 138 words per minute for Arabic (Trauzettel-Klosinski & Dietz, 2012). For Korean, which those texts do not cover, that bound is applied to eojeol counts and is therefore the least binding of the fixed bounds, and the size of the subset it defines will be reported. The premise is that a narration recorded for children is paced no faster than adult oral reading in the same language. If the selectivity survives where the recording was not the constraint, pacing does not explain it. Hypothesis 4 is reported as underpowered from the subset if the subset holds fewer distinct texts than the adjusted requirement. Should the data controller supply the per-text audio duration, or disclose a fixed narration rate for the engine, these rules are replaced by that quantity and we will report the substitution as a deviation from this protocol.

The Hypothesis 5 model will have the same fixed and grouping structure and a cumulative-logit likelihood over the three rating categories, with category thresholds allowed to vary by language. Language differences in use of the middle category are then not read as differences in text difficulty. Whether the Account intercept is estimable in this model depends on how many accounts contribute more than one rated reading, which the tier-1 count does not settle, so we will report that number and, where the intercept is not estimable, name child-level scale use as a limit on the rating. We will also report the band-specific odds ratios derived from the component-by-age-band terms beside the pooled ones, as descriptive quantities. A text rated by fewer than five distinct children does not contribute, and the primary model is complete-case, with the inverse-probability sensitivity analysis registered under the audit.

For tier 2, we will refit the same specification with a varying intercept for Account, decomposing accumulated reading into the account's mean and the current session's deviation from it (Curran & Bauer, 2011), alongside session order, days since the previous session and days since the account's first eligible session. The deviation term is the registered test for Hypothesis 6.

Posterior predictive checks will assess calibration and residual structure by language, age band, reading level and narration state in every model.

Priors

Priors for each confirmatory main effect, and for the secondary text predictors for which the review holds at least two studies, will be derived from the accompanying systematic review through a deterministic pipeline. We froze its rules before any extraction row existed, and will run it, archiving its output, before any outcome datum is accessed. The review's extracted estimates for the

corresponding predictor family and outcome class will be sign-harmonised so that facilitation always means faster reading, converted to a common standardised scale, and pooled by random-effects meta-analysis with one estimate per study per relation. The prior is Student's $t(3, \hat{\mu}, \sigma)$, where $\hat{\mu}$ is the pooled estimate translated to the model's scale and σ is twice the combined pooling and between-study uncertainty, with a floor of 0.10. The doubling and the floor guard against the gap between a laboratory comprehension test and a passively timed reading in a game.

The same review motivates the hypotheses, so the evidence that generates a prediction also centres its prior. Inference here is estimation against a region, and not a Bayes factor, so a prior centred in the predicted direction does not enter the decision as a hypothesis weight. And we will report prior-data conflict diagnostics for every review-derived prior. We register the tie-break as well: the fit with the derived priors carries the registered verdict, and a hypothesis whose label changes under the weakly informative refit described next is reported as sensitive to the prior and is not counted as a determinate confirmatory verdict.

Coefficients without adequate review evidence, which means every interaction term and any family supported by fewer than two studies, keep weakly informative priors: $t(3, 0, 0.5)$ for standardised main effects and $t(3, 0, 0.25)$ for interactions. Group-level and residual standard deviations take half-$t(3, 0, 1)$ priors and any estimated correlation matrix takes LKJ($\eta = 2$). Intercept priors will be calibrated by prior-predictive simulation to the outcome scale, without reference to any predictor-outcome association. The derivation rules are summarised in Supplementary Section S6, and the full prior table with its forest plots will be archived before the extract is transferred and will accompany the Stage 2 report. We will refit every model with weakly informative priors throughout, and with the derived scales halved and doubled, and report conclusions under each.

Decision rules and diagnostics

For every confirmatory coefficient, we will report the posterior median, the 95 per cent highest-density interval, the probability of the predicted direction, the proportion of the posterior inside the region of practical equivalence, and a model-implied contrast on an interpretable scale. The region is $|\beta| < 0.05$ standard deviations for a one-standard-deviation change in a predictor of reading time, and a half-width of log(1.05) on the log-odds scale for the rating model. More than 95 per cent of the posterior inside the region is read as practical equivalence, and less than 5 per cent inside it as a practically non-negligible association, labelled in the predicted or opposite direction by the sign of the posterior median. Everything else is inconclusive. For the ordering clause of Hypothesis 5, a posterior probability of at least 0.90 that the three coefficients rank as they do for reading time counts as agreement on order, and anything lower as order not established. For Hypothesis 4, at least 95 per cent of the posterior of the difference of attenuations above 0.20 supports selective removal of decoding cost. More than 95 per cent inside the region supports audio pacing, with the duration-adjusted contrast reported beside it. More than 95 per cent below -0.05 is consistent either with reverse selectivity or with pacing acting through the recording's duration, which the duration-adjusted contrast separates. Anything else is inconclusive. The fixed-denominator sensitivity contrast is read against the same thresholds.

Each hypothesis is a single pre-registered test with its own decision rule, and no claim in this protocol aggregates evidence across hypotheses, so no correction for multiple comparisons will be applied. The one composite claim, the overall Hypothesis 1 verdict, is an intersection-union rule: it is recorded only when all three coefficients are individually non-negligible in the predicted

direction, so its error rate cannot exceed that of any single component. Hierarchical shrinkage and the null condition of the design analysis, which reports how often a zero effect is wrongly declared non-negligible, calibrate the per-decision error rates behind Table 1.

Four registered contingencies read fitted quantities after fitting, and none of them changes whether a hypothesis is tested, on which sample, or with which model. The first is the Hypothesis 4 compression diagnostic, which withholds the full-sample contrast and, under the rule stated with Hypothesis 1, moves a Hypothesis 1 verdict to the derived unnarrated slope where the unnarrated readings cover that component's threshold. The second is the exclusion of draws with an unnarrated slope inside the region from the Hypothesis 4 ratio. The third is the third outcome-neutral check, which makes the within-level estimate primary for a component whose pooled coefficient it halves. The fourth is the prior-sensitivity label above. Each is stated where it applies, as an IF-THEN rule fixed here.

We will run four chains of 4,000 iterations, half of them warm-up. Acceptable sampling requires $\hat{R}$ below 1.01, bulk and tail effective sample sizes above 1,000 for every confirmatory parameter, no divergent transitions, satisfactory energy diagnostics and no unresolved maximum-treedepth warnings. Models failing these will be refitted with a higher target acceptance rate, greater tree depth and more iterations. If simplification is still needed, language-level slopes will be removed in the fixed order lexical, then discourse, then grammatical, and every estimable varying intercept will be retained. That order follows the expectations set out in the Introduction. Every simplification and its diagnostic consequence will be reported. Variance inflation factors will be reported for the fixed-effect design matrix. A high value does not trigger an outcome-dependent deletion: the joint model remains primary, and single-family models will be reported as a collinearity sensitivity analysis.

For Hypotheses 2, 3 and 4, we will compare the model with one omitting the relevant interaction, using Pareto-smoothed importance-sampling leave-one-out cross-validation, and report the expected log pointwise predictive density difference, its standard error and the Pareto-k diagnostics, with exact refits where importance sampling is unreliable. Predictive comparison supplements the coefficient-level summaries and does not gate them.

Outcome-neutral quality checks

Six checks are registered, each with a stated pass criterion and a stated consequence of failure. All but the fifth run before the confirmatory models are interpreted. The fifth is the ordering clause of Hypothesis 5 and is evaluated with that hypothesis. The first, second, fourth and sixth are decided without reference to any confirmatory coefficient, and the third is a robustness condition computed from the fitted text coefficients.

The first is a permutation null. We will reassign texts at random to readings within language and reading level, and refit the tier-1 model 200 times. The refits will use the maximum-likelihood analogue of the tier-1 model, with the same fixed and estimable grouping structure, and the region will be applied to the 95 per cent Wald interval, since 200 full Bayesian refits are not feasible and the check concerns leakage from the pipeline. No text coefficient may be declared practically non-negligible in more than 5 per cent of those refits. An inconclusive verdict on permuted data is the expected outcome at moderate text counts and does not count as failure. A coefficient that survives permutation means the pipeline is carrying structure that does not come from the pairing

of a text with a reading, and we will withhold the confirmatory results until the leak is found and reported.

The second is a positive control on the timing column. Total elapsed time must rise with the story's word count, with more than 99 per cent of the posterior above zero. Failure means the timing column does not index reading, and Hypotheses 1 to 4 and 6 are reported as not evaluable on this extract.

The third asks whether the association is an artefact of platform assignment. We will re-estimate the three text coefficients within each reading level separately, and each must retain at least half the magnitude it had in the pooled model. If a coefficient falls below half its pooled magnitude, the within-level estimate becomes the primary estimate for that component and the pooled one is reported beside it, with the assignment confound named wherever the coefficient appears.

The fourth is a set of three independence claims the theory object entails. Conditional on the reading level and on whether a text was generated or classic, the three components should be close to independent across texts. Narration should be unrelated to accumulated reading. And the difficulty rating should carry no further information about reading level once reading time is conditioned on. We will report each as an estimate with an interval. An implication fails when the 95 per cent interval of its conditional association lies entirely beyond 0.10 in standardised units. A failure does not invalidate the confirmatory tests, but it identifies a path the causal diagram omits, and it is reported as such. Failure of the third would identify a direct path to the reported difficulty from the reader's capacity, which the reading level partly indexes, or from the generation route. That path would then be named beside the Hypothesis 5 odds ratios. The criterion for failure is unchanged.

The fifth is convergent validity, the ordering clause of Hypothesis 5, specified there and evaluated with it, with its outcome under an underpowered Hypothesis 5 stated there.

The sixth is a validation of the feature pipeline itself. We will check a stratified random sample of 50 texts per language by hand for tokenisation, sentence splitting and dependency attachment, and report the agreement statistics per language before any feature is joined to an outcome. A language whose labelled-attachment agreement with the hand-checked parse falls below .85 contributes to descriptive reporting but not to the grammatical component.

Sensitivity analyses

Eight families of sensitivity analysis are registered, and every one of them is descriptive. Each bounds a primary estimate; none replaces it, and none changes which hypothesis is tested or how its verdict is reached.

The first family varies the outcome and the bounds placed on it. We will refit the timing models with no plausibility bounds at all, with winsorisation at the 1st and 99th and at the 0.5th and 99.5th percentiles within language, and with posterior-predictive residual trimming. Where the censoring rule is triggered, we will report the censored and uncensored fits side by side. We will also refit the tier-1 model with the outcome expressed per character, reporting the three component coefficients beside the primary estimates.

The second varies what counts as an eligible reading and an eligible text. We will move the tier-2 threshold from three distinct texts to two and to five, and add repeated readings with a repetition indicator. We will refit the tier-1 model with the sentence floor raised to eight, reporting the three component coefficients and the distinct-text count beside the primary estimates. We will also

compare Text grouping with grouping on the attribute combination that generated the text. Because the classic stories are likely to be public-domain texts that the surprisal model's training corpus could contain, we will report the mean per-word surprisal of classic and of generated texts per language before any outcome is joined, and refit the tier-1 model on generated texts alone, reporting the three component coefficients beside the primary estimates.

The third concerns language. For Hypothesis 6, we will refit the tier-2 model with an uncorrelated language-varying slope on the accumulated-reading deviation, where Language meets the grouping-factor estimability rule within tier 2, and report the per-language slopes descriptively; that refit is dropped, with the drop reported, if it fails the sampling diagnostics registered under Decision rules and diagnostics. We will restrict analyses to the five largest languages, and fit language-specific models and component solutions with loading congruence reported. From those fits, we will report the two age-band interactions for each language beside the pooled coefficients, arranged by orthographic depth where a published classification places the language, and we will refit the tier-1 model with the two age-band interactions varying by Language. The design analysis priced neither of the last two, per-language posteriors will be wide, and neither gates any verdict, changes the primacy of the pooled coefficient, or supplies a second test of Hypothesis 2 or 3. The language-varying interaction slopes sit outside the tier-1 specification, so the simplification order registered under Decision rules and diagnostics does not remove them.

The fourth asks how much the estimates depend on who is reading in which language. Before the transfer, we will deposit a lookup table marking whether each reading language is an official or majority language of each country, as a coarse indicator of the match between the two. We will report the cross-tabulation of reading language by account country and the number of accounts reading in more than one language, and then refit the tier-1 model on sessions whose reading language is a language of the account's country, reporting the three component coefficients and the subsample's distinct-text count beside the full-sample estimates and the threshold in Table 1. The indicator enters no confirmatory model. It misclassifies in both directions, so the refit bounds the estimates' sensitivity to sample composition and identifies no individual reader.

The fifth varies how the text properties are scored. We will replace the component scores with sparse-component scores and with pre-specified equal-weight composites of the anchor features. For Hypothesis 3, we will also report the age-band interaction of each ReferentialContinuity anchor from a model in which that anchor replaces the component, as a check on whether the three devices agree in sign.

The sixth varies the age contrast. Where the audit finds the age field finer than two bands, we will re-estimate the Hypothesis 2 and 3 interactions with the finer age entered in two ways, as an ordinal monotonic predictor (Bürkner & Charpentier, 2020) and as an unordered factor with each finer band contrasted against the 7-8 band, and report both beside the two-band contrast. We will also refit the tier-1 model with pages per hundred words, computed from the recorded page count, and its interaction with the age band, reporting the Hypothesis 2 and 3 interactions from that fit beside the primary ones, as a check on how much of the age difference is time away from the text.

The seventh concerns the two hypotheses with an outcome of their own. For narration, we will report the within-account contrast where it is not already primary, the fixed-denominator contrast registered under Confirmatory models, the duration-adjusted contrast from the refit that carries the

predicted spoken duration per word and its interaction with narration, and the Hypothesis 4 contrast computed within each age band. For the rating, we will compare the model under adjacent-category and cumulative-logit links, stratify it by language, and compare the inverse-probability analysis for rating non-response with the complete-case estimates.

The eighth varies the region itself. We will apply the alternative regions of practical equivalence, 0.03 and 0.10 standard deviations for reading time and half-widths of log(1.03) and log(1.10) on the log-odds scale for the rating.

To ease comparison with the prior literature, we will report maximum-likelihood mixed models with the same fixed and estimable grouping structure in a supplementary table. They are descriptive and will not be used to resolve a Bayesian result.

What is not yet settled, and what follows from each answer

We put thirteen questions about the extract to the data controller on 7 August 2026, and they remain open at submission. Table 2 attaches a decision rule to each in advance, and every rule below is a function of the answer alone. Five of these questions correspond to auxiliary assumptions recorded in the theory object.

Table 2

Questions outstanding at submission and the decision rule attached to each

What is unresolved	What it bears on	What the protocol does under each answer
How many distinct texts carry eligible readings, settled by the provider-computed counts together with question d. The controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes yields one shared text or a fresh text per session. The published pre-launch description is a catalogue of over 40,000 generated stories.	Every text coefficient, since all are identified between texts.	The count will be read from the provider-computed figures before modelling and located on the precision curve in Figure 3. Below the threshold in Table 1 a hypothesis is declared underpowered before its model is fitted. Where each session carries its own text, the Text varying intercept is replaced by varying intercepts for the repeated attribute combination, under the sparsity rule given under Measures, and precision is then governed by the number of attribute combinations. We will report which case obtains.
Whether the narration flag records that audio played or only that it was available, and whether it can change within a session, together with whether the voice is recorded or synthesised, how many voices serve a language, whether words or lines are highlighted in time with the voice, whether a page advances automatically when playback ends, whether playback can be paused or repeated, the rate at which the engine plays, and the per-text audio duration (question a).	Hypothesis 4, through the auxiliary assumption that narration played.	If it records playback, Hypothesis 4 stands as written. If it records availability only, the variable is relabelled as availability, the contrast becomes an intention-to-treat comparison whose attenuation is diluted towards the null by non-compliance, and that dilution is stated wherever the coefficient is reported. Sessions in which the flag changes are excluded and counted. If pages advance automatically when playback ends, narrated sessions are the case the compression diagnostic registered under Confirmatory models is designed to detect, and that diagnostic's registered consequence applies unchanged: the full-sample contrast is not interpreted and Hypothesis 4 rests on the unpaced subset. If words or lines are highlighted in time with the voice, that is reported as a boundary condition wherever a narration coefficient appears, and no coefficient changes. If the engine's rate setting or the per-text audio duration is disclosed, the substitution already registered in the third Hypothesis 4 rule applies.
Whether one story identifier maps to one text, and whether a text differs across reading languages by translation or by independent generation (question d). The published pre-launch description says many stories were translated from English.	The identity of the stimulus, through the auxiliary assumption that the keyed text is the text the child read, and therefore every model's grouping structure.	If one identifier is one text, texts are counted by identifier crossed with language and reading level, as specified. If a text is translated across languages, a varying intercept for the underlying story is added, crossed with Language. If each language is generated independently, texts in different languages are distinct and no such term is added.

What is unresolved	What it bears on	What the protocol does under each answer
Whether a stable identifier exists for each session (question e). The sample carries none.	The join between the extract and the texts, and the check of the transferred file against its manifest.	If one exists, it is carried in the export and used for the join. If none exists, sessions are joined to texts on story identifier, reading language, reading level and app version, as Data provenance states, and the transferred file is checked against its manifest by row count and checksum. Neither answer changes a model.
What the reading-time interval covers: whether it includes embedded activities, and whether it includes time backgrounded or idle (question c).	The outcome itself, through the auxiliary assumption that elapsed time indexes effort.	If the interval excludes them, the outcome stands as written. If it includes embedded activities, texts that offer them are identified from the story attributes, an indicator enters every model, and the primary analysis is repeated within texts that offer none. If it includes idle or backgrounded time, the upper plausibility bound and the censoring rule carry more of the work, and the limitation is stated in the abstract itself.
Whether an account persists across reinstalls and whether siblings can share one (question f).	Tier 2 and Hypothesis 6, through the auxiliary assumption that an account is a child.	If an account can be shared, the within-child reading of Hypothesis 6 is withdrawn in advance and the coefficient is reported as a within-account association, with the ambiguity stated in the hypothesis itself.
Whether the reading level can change within an account over time, and whether it changes the text served or only the page count (question g).	The platform-assignment confound and the third quality check.	If it changes only pagination, texts are comparable across levels and the pooled estimate is primary with the within-level estimate as a check. If it changes the text, the within-level estimate becomes primary and the pooled one is reported alongside it. A within-account contrast on a level change is not registered here, and any such analysis would appear at Stage 2 under Exploratory analyses.
The source of the age band and how much of it is missing across the whole window (question h). It was blank for 10 of 35 sessions in the structural sample.	Hypotheses 2 and 3.	Missingness will be reported by language, reading level and app version. Above 20 per cent, availability is modelled from observed pre-response variables and an inverse-probability-weighted fit is reported alongside the complete-case fit, with both interpreted together.
Which reading languages the window covers. The app is marketed in 19 languages and the structural sample showed ten (question i).	Coverage of the feature pipeline and the language grouping factor.	The observed set will be reported as found. A language for which the required resources do not exist contributes to the participant-flow diagram and the descriptive tables but not to any feature-dependent model, and the excluded volume is quantified.
Whether the classic stories are fixed, human-authored texts identical for all users within a language (question j).	The comparison between machine-generated and human-authored material, contingent hypothesis C4.	The condition was not confirmed before submission, so C4 is not activated. The generated-or-classic indicator remains a covariate, and if the condition is confirmed later the comparison would appear at Stage 2 under Exploratory analyses.
Whether unfinished sessions are retained anywhere, and what the completion rate is (question k).	Generalisability, since every row in the extract is a completed reading.	The study describes completed readings and says so. Where an aggregate count of unfinished sessions by language is available, the completion rate is reported and the restriction quantified.

What is unresolved	What it bears on	What the protocol does under each answer
What fields the older data carry and what window they cover (question l).	Hypothesis 6, through the within-window assumption that accumulated reading acts on decoding automaticity over the export window and not the reverse. The measure is truncated by a window opening on 1 January 2026.	If an older file carries the same fields, accumulated reading is computed over the longer window and the analysis is repeated on both. Otherwise the measure is left-truncated at the window's opening and is described throughout as reading accumulated within the observation window, which makes an experienced reader look new.
Whether post-story responses include comprehension items or only the difficulty rating (question m).	Contingent hypotheses C1 to C3.	The condition was not confirmed before submission, so C1 to C3 are not activated. If item-level responses turn out to exist, the analyses written for them would appear at Stage 2 under Exploratory analyses, and no comprehension claim is made among the confirmatory results.
What the per-session activity counters measure. In 6 of the 14 structural-sample sessions with activity, they sum to more than the whole reading duration, in one case to nearly twice it (question b).	Nothing, by decision.	These fields are set aside and are not used as a predictor, a covariate or an exclusion criterion. The decision is recorded here so that their later use would be visible as a deviation.

Contingent hypotheses

We specify four further hypotheses in full here, outside the confirmatory set, because each depends on a fact about the extract that the data controller has not yet confirmed. Each carries an activation rule that rests on that fact alone, read from the controller's written answer or, for the requirement that accounts be observed in both narration states, from the provider-computed counts.

A contingent hypothesis activates only if its condition is confirmed in writing by the data controller before this protocol is submitted, and the confirmation is reproduced in Supplementary Section S7. A condition confirmed after submission does not activate anything: the hypothesis is then reported at Stage 2 as exploratory and is not counted among the confirmatory tests. At submission, no condition has been confirmed, so none of the four is activated, and Supplementary Section S7 records that. They are retained here so that, should a condition be confirmed later, the analysis reported at Stage 2 under Exploratory analyses is the one written before any data were seen. A hypothesis whose condition is never confirmed is not run.

C1: coherence and inferential comprehension. Condition: item-level comprehension responses exist and carry a recoverable question type. We would predict that, in an item-level model of response correctness, ReferentialContinuity × QuestionType is positive with inference questions coded 1 and literal questions 0, so that the association of continuity with correctness is larger where the question requires integration. The prediction is assembled from three separate results rather than taken from any one of them: an inference question is answered by integrating information across clauses (Freed & Cain, 2021), cohesion predicts children's comprehension overall (Pickren et al., 2022), and text properties bear differently on surface and inferential questions (Aparicio et al., 2022). We located no study testing a continuity by question-type interaction directly, which is one reason this hypothesis is contingent.

C2: narration and comprehension. Condition: item-level comprehension responses exist, and accounts are observed both with and without narration. We would predict that, within an account, narration is positively associated with response correctness after text, age band, reading level, accumulated reading and language are adjusted for. The expected size is small (Clinton-Lisell, 2023; Wood et al., 2018), and the coefficient identifies an association, and never an effect.

C3: reading pace and comprehension. Condition: as C2, restricted to sessions without narration. We would predict that an account's mean log reading time per word is negatively associated with response correctness, so that accounts with a faster habitual pace answer more accurately (Sortkær et al., 2021). The simultaneous within-account deviation term keeps this between-account association distinct from a difficult or interrupted session.

C4: machine-generated against human-authored text. Condition: the classic stories are confirmed to be fixed, human-authored texts, identical for all users within a language. We would predict that the three text components predict reading pace with coefficients of the same sign in generated and in classic texts, and that the interaction of each component with the generated-or-classic indicator is practically equivalent to zero under the registered region. This is a test of exchangeability between the two kinds of text, and the closest existing evidence is the adult eye-movement corpus on machine-generated text (Bolliger et al., 2025). A non-negligible interaction would mean that the properties governing how children read human-written stories govern their reading of generated ones differently.

Timeline

The export is built and held before submission, so the study can begin on the day in-principle acceptance is confirmed. We anticipate transfer of the extract within four weeks of that confirmation. The frozen audit, the feature extraction with its hand-checked validation and the outcome-blind dimension reduction will follow within three months of transfer, and the confirmatory and sensitivity models within a further four months. The Stage 2 manuscript will be submitted within twelve months of in-principle acceptance. [The author to confirm or adjust these intervals.] Any extension would be negotiated with the handling editor.

Data availability

The session-level extract is held by the data controller and is not in our possession at submission. On in-principle acceptance, it will be transferred under a data-sharing agreement between the University of Oxford and the data controller. On acceptance of the Stage 2 manuscript we will deposit, in [the named public repository, with a digital object identifier], the story texts, the computed text features, the component scores, the fitted model objects, the participant-flow counts, the laboratory log, and every table and figure underlying the reported results. Session-level fields that the agreement classes as commercially sensitive may be restricted. For each such field, we will state the reason and deposit a derived form sufficient to reproduce every reported analysis, and interested researchers will be able to request the restricted fields through [the named data controller contact or the mechanism the agreement specifies]. The file manifest and the provider-computed aggregate counts contain no record of any individual and will be deposited in the same repository on receipt, before in-principle acceptance. The 35-row structural sample described under Pilot data is deposited there for peer review, under embargo until Stage 2 acceptance, with a view-only link supplied to the editors.

Code availability

All analysis code is version-controlled at https://github.com/pablobernabeu/InTRACt_Interplay_of_Text_Reader_And_Context, in the `preregistration` directory, which will be public and complete at submission and is available for peer review. The theory object with its checks and derived diagram, the design specifications with the precision analysis and its re-solve at 0.95 assurance, the count script supplied to the data controller, and the literature sizing are deposited there, so the theory object, the design analysis and the literature sizing can be reproduced now. The outcome-model code will be written against simulated data, as described above, and deposited in the same location before the extract is transferred, together with the frozen feature inventory and the archived prior table. The four packages the protocol relies on are cited by their concept identifiers, and the releases in use are theoryforge 0.6.0 (Bernabeu, 2026d), pilotr 0.3.0 (Bernabeu, 2026b), scopusflow 0.4.0 (Bernabeu, 2026c) and depictr 0.3.0 (Bernabeu, 2026a). The software versions behind the reported numbers are recorded beside their outputs, in `design_analysis/session_info.txt` and `design_analysis/session_info_ordinal.txt` for the design analysis, `theory/tf_check.txt` for the theory object and `literature/scopus_run_stamp.txt` for the literature sizing. The design-analysis figures were composed with depictr, whose audit checks each panel for a floor on rendered type size and for separation in greyscale at the width the journal prints, and refuses to save a panel that fails.

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Figures

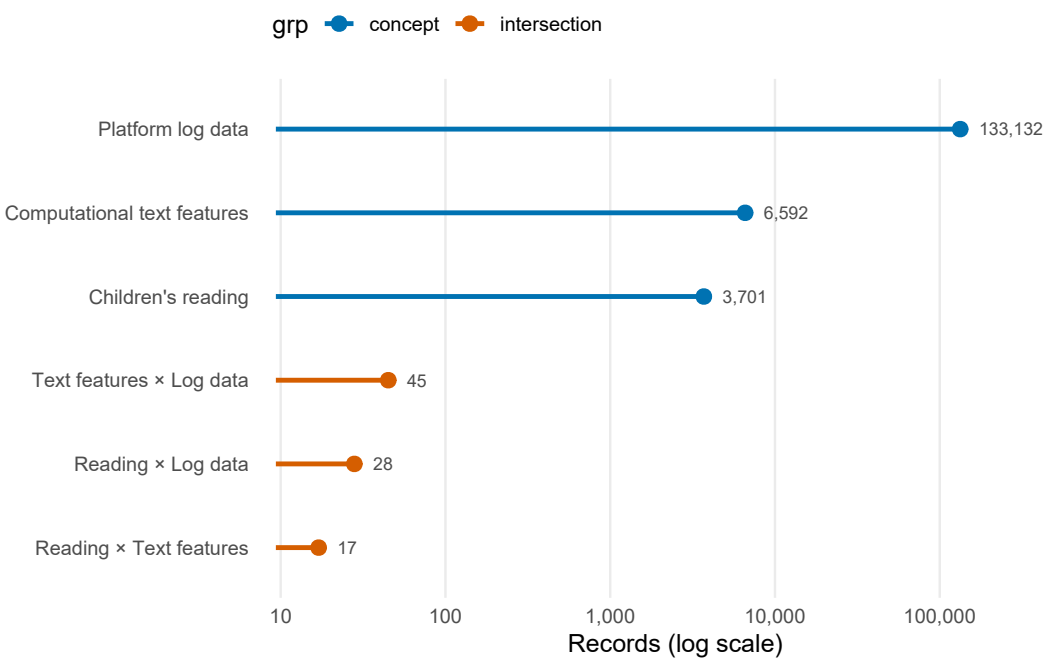


Figure 1

Sizing the literature around this study. Records on Scopus from 2018 to the search date of 26 August 2026, for children's reading comprehension, computational text features and platform log data, and for every intersection of the three. The pairwise intersections hold 17, 28 and 45 records, and the three-way intersection holds none, so it does not appear on the logarithmic axis. Search expressions and the date of the search are in the supplementary record.

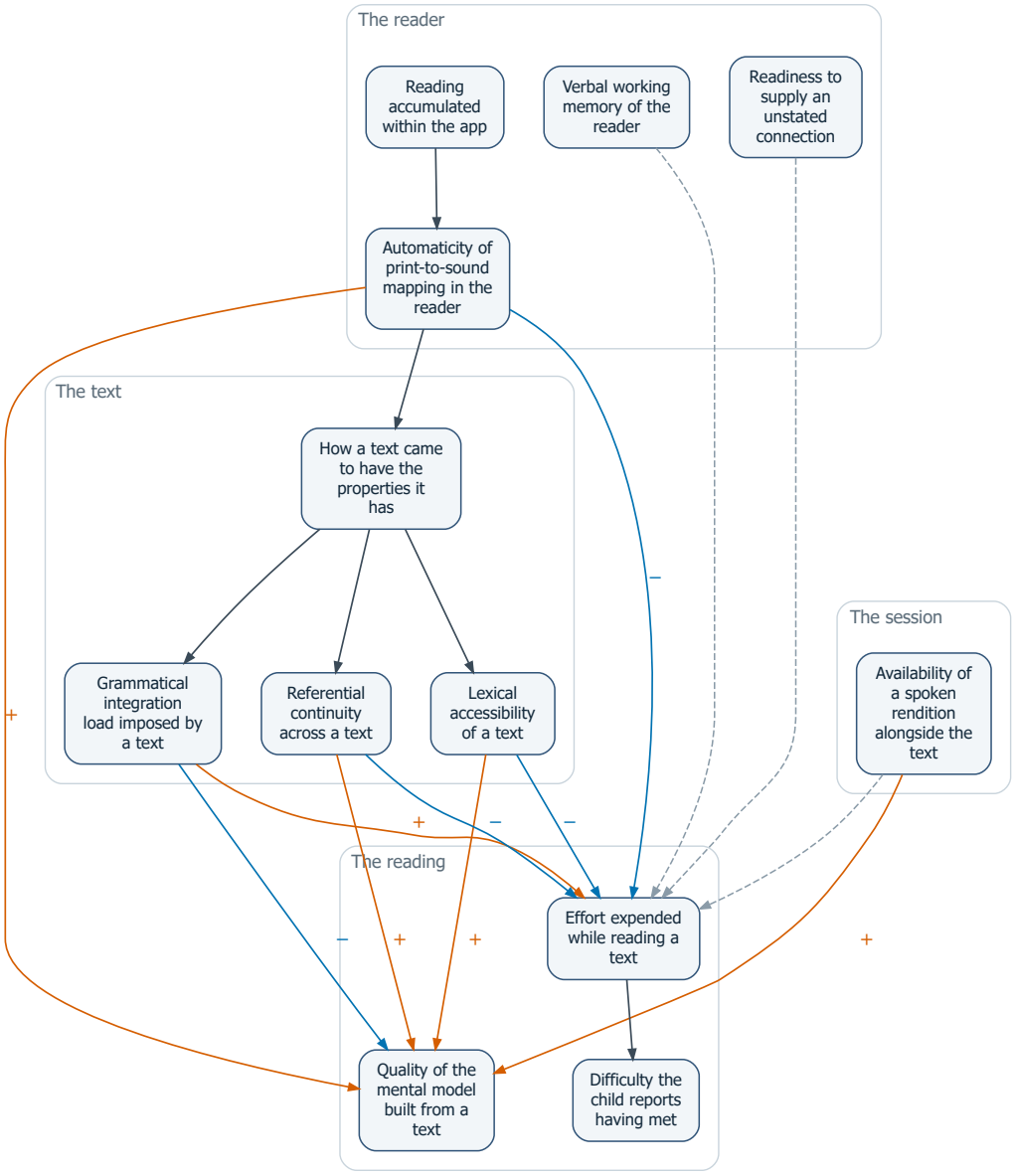


Figure 2

The nomological net, generated from the deposited theory object. Solid arrows are causal paths, with the predicted sign shown on the line where the theory fixes one, and dashed lines mark moderation. Text properties and reader capacities converge on the effort a reading costs, which shapes the difficulty the child reports. A parallel set of paths reaches the quality of the mental model built, which these data can measure only if comprehension items prove to exist. How a text came to have its properties, meaning the generation route and the reading level being served, is a common cause of all three text properties and receives a path from the reader's own decoding automaticity, which is why every text coefficient is estimated within reading level.

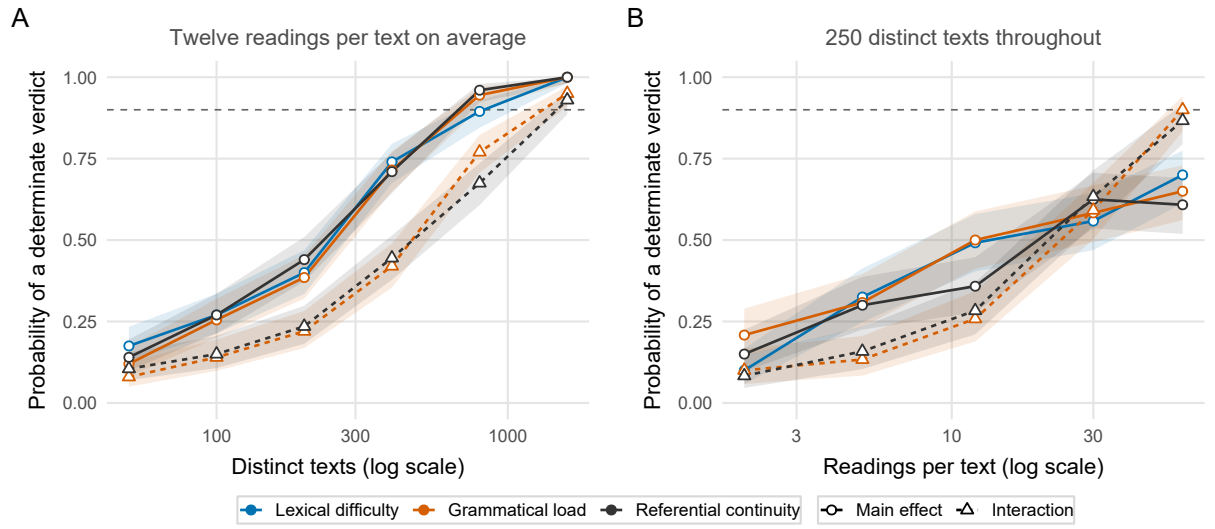


Figure 3

Precision as a function of what there is more of. Panel a sweeps the number of distinct texts at an average of twelve readings each, and panel b holds the number of texts at 250 and sweeps readings per text. Points are the probability that a 95 per cent interval falls determinately outside the region of practical equivalence for an effect of 0.10 standard deviations, over 200 simulation replicates per grid point in a and 120 in b. Shaded bands are 95 per cent Wilson score intervals on each probability, and the dashed line marks the 0.90 target. The two panels answer different questions, and only one of them is about having more data of the kind this platform generates most readily.